Financing Social Security Reform without Losses

HongSeok Kim*

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Abstract

I construct a Social Security transition that shares the gains from a smaller pay-as-you-go pension across generations using two compensation instruments. I study a general-equilibrium overlapping-generations model calibrated to the 2010 United States in which work builds skills and the government faces a debt ceiling. The reform combines a gradual payroll-tax reduction with cohort-specific payroll relief during working life and benefit supplements in retirement. General revenue finances the compensation, with a consumption tax balancing the budget. I find that a Pareto-improving transition across birth cohorts can be achieved without a broader income-tax reform. Financing cohort-specific payroll relief and retirement-benefit supplements raises the consumption tax, yet leaves every evaluated cohort at least as well off on average. The key implication is that protecting transition generations need not require preserving pension promises or overhauling the tax system: a set of contribution and benefit adjustments can suffice, even after accounting for the cost of financing compensation.

Keywords: Pareto-improving transition, intergenerational redistribution, debt ceiling, learning by doing.

JEL: H55, H63, E62, D15, J26.

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1 Introduction

In 2034 the Social Security trust fund is projected to be depleted,[1] and the payroll tax that remains will cover about 83% of the benefits now scheduled (Social Security Trustees, 2026). Congress will then have to raise taxes, cut benefits, or borrow, and each of these choices has the same problem: someone alive today pays for it, either in higher taxes during their working years or in smaller checks during retirement. A smaller program would raise output and wages in the long run, so the people who are not yet born would gain while the people who are alive now would carry the cost of getting there. This paper asks whether the gains from a smaller Social Security program can be delivered without leaving any birth cohort, present or future, worse off on average than it would be if nothing were done, what it takes to pay for that, and how large a program the transition should land on.

The literature has studied this question for decades, and the answer has mostly been no. A pay-as-you-go pension gives its first generation benefits it never paid for, and later generations earn a return equal to wage growth rather than the interest rate (Samuelson, 1958; Aaron, 1966). That initial windfall is a debt that someone must eventually carry, so moving toward a funded system cannot avoid making some generation worse off (Breyer, 1989; Geanakoplos et al., 1998; Sinn, 2000). Quantitative life-cycle models confirm this along the transition: future cohorts gain and the cohorts alive at the reform lose (Auerbach and Kotlikoff, 1987; Huang et al., 1997; Conesa and Krueger, 1999; Nishiyama and Smetters, 2007).

McGrattan and Prescott (2017) turned the question into one of transition design and showed that it can be answered: a phase-out of Social Security and Medicare, combined with a broader base and lower marginal rates for the income tax, leaves every cohort better off in their economy. I extend their question and give a different answer. I use Social Security's own instruments, cohort-specific contributions and benefits paid from general revenue, so no income-tax reform is needed to protect the transition. Endogenous human capital accumulation, the channel their analysis omits, turns out to be what makes the compensation affordable. And the same construction is applied at every long-run size of the program, including complete phase-out, so that whether a transition creates losses and how large a program to keep become two separate questions: the first is settled by the policy design, the second by social weights.

I study this question in a large-scale overlapping-generations model in the tradition of Auerbach and Kotlikoff (1987), with the demographic and fiscal structure of Braun and Joines (2015).

[1] The OASI and DI trust funds are legally separate. The combined projection assumes legislation that permits transfers between them. The calibration uses this combined projection.

Households enter at 20, face mortality risk, retire at 65 and live to at most 89, and the population ages along the Trustees' projected path. Households differ in assets and skills, value the estates they leave (De Nardi, 2004), and build skills through work (Imai and Keane, 2004; Chang et al., 2002), so the taxes that finance pensions change future earnings as well as current hours. The government runs Social Security and the Treasury as separate accounts, borrows up to a debt ceiling, and closes the remaining gap with a consumption tax, the financing margin used in Kitao (2014) and Braun and Joines (2015).

I discipline the model so that it reproduces the features of the U.S. economy that bear on a pension reform. Households save, work, and build skills at the rates observed in 2010. The economy holds three years of output in capital, adults work the hours recorded in the American Time Use Survey, and wages rise and then flatten over the working life as they do in the Current Population Survey, so that the response of saving, labor supply, and future earnings to a tax or benefit change is anchored to data. Wealth is as unequal within each age group as in the Survey of Consumer Finances, and the flow of estates matches what the wealth and mortality data imply, so the reform's effect on who inherits and who leaves nothing is also disciplined. On the fiscal side, the model starts from the observed public debt and trust-fund reserves, Social Security's benefits in payment and benefit formula, its payroll base, and government spending from the national accounts. The population then ages along the Trustees' projection, and the model reproduces the Trustees' reserve-exhaustion year and their projected program cost in 2100.

In the baseline, once the government can no longer borrow, benefits fall to what payroll revenue covers, which I refer to as a funding-linked benefit cuts. Shrinking the program instead creates resources: a lower payroll tax raises the return to work, more work builds skills and future earnings, and smaller benefits lead households to save more, so capital, wages, and output rise. These gains accrue mostly to cohorts who spend their working lives under the lower tax. The cohorts who lose are those who pay the full payroll tax for most of their careers and then retire into the reduced benefits. A reform that leaves no one worse off must therefore move part of the gain to these cohorts. If it finances that transfer with a consumption tax, the same retirees it means to protect pay part of the bill.

The reform follows from this pattern of losses. I announce the cut early but start it late enough that today's retirees and older workers are not affected, and I cut gradually so that each cohort's benefit loss is matched by payroll savings over its own working years. The cohorts caught between, who pay the old tax for part of their careers and retire into the lower benefits, I compensate with a lower payroll rate while they work and a higher benefit after they retire. The supplement is

paid late in life, so its cost falls in years when most taxpayers are themselves beneficiaries of the reform. General revenue pays for both instruments, so no cohort is singled out to finance another. Public debt stays at the same ceiling relative to GDP as in the baseline, so the compensation is paid out of current revenue, which the higher output makes possible, rather than by relaxing the fiscal rule. I measure each cohort's welfare as the population-weighted average of its members' lifetime utility and size the two instruments cohort by cohort until no birth cohort loses. To rank reforms, I aggregate cohort welfare with a social discount factor over those not yet born. I then apply the same construction to other long-run payroll rates, including a complete phase-out, and compare these compensated transitions with one another.

Each piece of this reform has been used in practice. Germany, Japan, Canada, and Australia all pay part or all of their public pensions from general revenue, and so does the United States for Medicare Parts B and D (Medicare Trustees, 2025). Pension reforms are routinely phased in by birth cohort, as the 1983 Social Security amendments did with the retirement age, and Chile compensated the cohorts caught in its 1981 transition with Treasury-paid recognition bonds. Since 2022 Canada pays Old Age Security recipients aged 75 and over a benefit ten percent higher than younger recipients receive. This paper combines these instruments into one transition and asks how large each must be, cohort by cohort, for no birth cohort to lose on average, and whether a government that holds its debt at a fixed ceiling can afford them after the consumption tax has adjusted.

The main finding is that such a transition exists and that it is not costly. Cutting the payroll tax from 12.4% to 3.5% between 2060 and 2200, with the two compensation instruments sized cohort by cohort, leaves none of the 804 birth cohorts I evaluate worse off on average than under the baseline, with ordinary earnings- and capital-income tax rates unchanged and the consumption tax adjusting to balance the budget. The largest payroll relief puts a cohort's payroll tax rate 2.3 percentage points below the economy-wide rate, at 10.1% instead of 12.4%. The largest late-life supplement raises a cohort's annual benefit by 8.6% above what it would otherwise receive. Summed over the transition and discounted, the Treasury's spending on the two instruments equals 2.2% of GDP in 2026. Because public debt stays at its ceiling, the consumption tax must raise this money. The consumption tax rate exceeds the baseline's by at most 0.45 percentage points while the relief is phased in, and from 2072 on it is lower, because output is larger while government spending outside Social Security grows with productivity rather than with output, and the interest rate on the public debt is lower. By 2200, capital per adult is 23% higher than in the baseline and consumption per adult 16% higher. A cohort born in 2100 gains the equivalent of

3.4% of lifetime consumption.

With the same instruments and compensation rule, I find a Pareto-improving transition to every long-run payroll rate from zero to 8%. A compensated phase-out to zero, for instance, leaves no cohort worse off and yields a long-run gain of 10.5% of lifetime consumption, against 7.7% at a 3.5% rate, at a Treasury cost of the same order. Among these transitions, the 3.5% rate gives the highest social welfare for discount factors between 0.83 and 0.91, so it is constrained optimal over that range, and lower rates do better for a planner who weights the unborn more heavily. Without this protection, a phase-out that starts cutting in 2026 and compensates no one leaves 51 cohorts worse off, and any of the compensated transitions is preferred to it unless the planner puts heavy weight on future generations.

For robustness, I test six changes and re-solve compensation in each case to protect every cohort. Removing learning by doing and recalibrating increases the number of losing cohorts under the uncompensated cut from 35 to 100. The long-run gain falls from 7.7% to 2.8% of lifetime consumption. With the two instruments, I cannot find a transition that protects every cohort. The gains are too small and arrive too late to finance compensation. Removing the bequest motive and recalibrating preserves the Pareto result and the comparison with the phase-out. This holds whether the assets of the deceased pass within cohorts or across age groups. When Social Security is the only annuity, the welfare comparison favors retention over a wider range of social discount factors. Removing the consumption floor does not change the result. Financing payroll relief with a surcharge on cohorts born in 2044 or later reduces social welfare only slightly. Temporary borrowing above the debt ceiling raises welfare slightly. Delaying completion of the tax cut to 2300 removes much of the gain. Protecting every cohort on average can still leave some households worse off. Under the main reform, 4.3% of adults alive in 2026 lose, by at most 0.17% of consumption. I therefore re-solve the two instruments using each cohort's worst-off household, and find that the ranking of the retained program and the phase-out remains unchanged.

Related Literature. The paper belongs to the literature on the transition cost of pension reform. Breyer (1989), Geanakoplos et al. (1998), and Sinn (2000) show that the windfall paid to the first generation of a pay-as-you-go system is an implicit debt that a move toward funding cannot erase, so some generation must carry it. Quantitative life-cycle models confirm the pattern along the transition: in Auerbach and Kotlikoff (1987), Huang et al. (1997), Conesa and Krueger (1999), and Nishiyama and Smetters (2007), future cohorts gain from privatization and the cohorts alive at the reform lose, and a Pareto improvement requires lump-sum transfers across cohorts of

the kind a legislature cannot write. Kitao (2014) compares the financing options, higher payroll taxes, lower benefits, a later retirement age, and a consumption tax, under a fixed fiscal rule, and Braun and Joines (2015) and İmrohoroğlu et al. (2016) place the problem in an aging economy with Medicare and public debt. McGrattan and Prescott (2017) come closest to this paper: in an economy with annuities and no skill formation, they find a phase-out of Social Security and Medicare that leaves every cohort better off, but only together with a broader income-tax reform. This paper keeps their question and changes the implementation in three ways. The compensation uses cohort-specific payroll relief and a late-life benefit supplement, financed through the general budget, with income-tax rates unchanged, rather than a reform of the tax base. The construction is applied at several long-run sizes of the program, including zero, and shows that the Pareto property is a feature of the transition design. And the model includes learning by doing, so the taxes that finance the benefit change future earnings as well as current hours. This channel makes the transition affordable: without it the long-run gains fall by two thirds and the same instruments no longer protect every cohort.

A second literature asks what size of Social Security is optimal. Positive answers rest on insurance: against longevity when annuity markets are missing, against earnings and health risk, and through redistribution across different types (İmrohoroğlu et al., 1995; Hubbard et al., 1995; Huggett and Ventura, 1999; Conesa and Krueger, 1999; Hosseini and Shourideh, 2019), with Conesa and Garriga (2008) and Peterman and Sager (2022) studying the optimal mix of social security, taxes, and public debt in life-cycle economies. Jang et al. (2025) show, with the steady-state welfare of a newborn behind the veil of ignorance as the criterion, that the zero-optimal-size result reflects the program's design: it is too distortive and not redistributive enough, and once the design is changed a positive or larger size is optimal. Earnings and health are deterministic here, and inheritances together with the bequest motive leave little longevity risk to insure, so the long-run optimum in this economy is zero, as the Aaron condition predicts when the interest rate exceeds the growth rate (Aaron, 1966). The paper isolates the transition component of the optimal size: with a no-losers constraint and implementable instruments, the constrained-optimal rate is positive over a range of social discount factors because a shallower cut leaves more cohorts with gains of their own instead of compensation. The insurance component would add to that level.

Two features of the model have not been studied together, and the question needs both. Learning by doing (Imai and Keane, 2004; Chang et al., 2002; Wallenius, 2011) decides how large the gains from a lower payroll tax are and when they arrive: more work raises skills and future earnings, so the tax base grows by more, and later, than the hours response alone would deliver,

and Blandin (2021) and Peterman (2016) show that this changes the welfare effects of payroll and income taxes. The debt ceiling, enforced by a consumption tax (Bohn, 1998; Altig et al., 2001; Hansen and İmrohoroğlu, 2016; İmrohoroğlu et al., 2016), decides whether those gains can be used to protect the transition: measured against each policy's own GDP, it forces the compensation to be paid from the revenue the reform itself generates, and the consumption tax that pays it falls partly on the retirees it protects. The literature has studied the two apart, skill formation in models of tax reform without a constraint on public borrowing and debt ceilings in aging economies without skill formation. Whether a transition can be financed without losers depends on both: the gains must arrive early and large enough, through skills, to cover the compensation under a rule that forbids borrowing against them. Without the ceiling any transition could be made Pareto improving by borrowing against future output; without learning by doing the gains come too late. Borrowing is costly here because the interest rate exceeds the growth rate throughout, unlike the low-rate settings of Blanchard (2019), and Auclert et al. (2021).

Outline. Section 2 presents the model and Section 3 its calibration. Section 4 describes the baseline economy, constructs the transition, and reports its welfare and macroeconomic effects. Section 5 compares the retained program with a complete phase-out and across long-run payroll rates. Section 6 examines the bequest motive, the financing of the compensation, and the borrowing allowance. Section 7 concludes.

2 Model

The model is a general-equilibrium overlapping-generations economy based on Auerbach and Kotlikoff (1987). Its demographic and fiscal components are similar to those in Braun and Joines (2015). Each model period is one year.

Households enter the model at age 20. They retire at age 65 and die by age 90. Households differ in their assets and skills. They choose consumption, saving, and hours of work. They cannot borrow. Their skills increase through work. Households also value the bequests they leave. Mortality is their only source of idiosyncratic risk. Firms use capital and efficiency units of labor to produce one good. The government has four components. Social Security pays benefits from payroll taxes and its trust fund. Medicare and a social safety net provide support for the elderly. The Treasury uses a consumption tax to balance the overall government budget. Households have perfect foresight along deterministic policy paths.

2.1 Demographics and productivity

Let $N_{j,t}$ be the population of age j at the start of year t, and $\pi_{j,t}$ its probability of surviving that year. Within each calendar year, people of the same age face the same survival probability. Both series use historical estimates and the intermediate projection of the 2026 Trustees Report through 2100 (Appendix Figure A.2). After 2100, I use the Markov population process of Braun and Joines (2015) to extend the population to a stationary distribution.

I measure population relative to $\overline{N}^W$, the number of adults aged 20–64 in 2010:

$$n_{j,t} = \frac{N_{j,t}}{\overline{N}^W}, \qquad n_t^W = \sum_{j=20}^{64} n_{j,t}, \tag{1}$$

where n_t^W is the working-age population in these units.

The old-age dependency ratio is the number of adults aged 65–89 per 100 adults aged 20–64. This ratio approximately doubles between 2010 and the stationary population (Figure A.1). Labor-augmenting productivity A_t grows at a constant rate g. I normalize it so that $A_{2010} = 1$. I express quantities in 2010 dollars. For computation, I divide quantities by A_t so that this gives a balanced-growth path in the long run.

2.2 Environment

A household of age j in year t starts the year with assets $a \geq 0$ and skill $h > 0$. It chooses consumption c, saving $x \geq 0$, and hours of work $\ell \in [0, 1)$. Hours are measured as a fraction of sixteen hours per day. Households must retire at age 65, so $\ell = 0$ from that age. Households adjust labor supply only through hours of work before age 65. Following Braun and Joines (2015), households face no idiosyncratic earnings risk.[2]

Households value consumption and leisure through a constant-relative-risk-aversion function of a Cobb–Douglas composite of the two:

$$u(c, \ell) = \frac{[c(1-\ell)^v]^{1-\sigma} - 1}{1-\sigma}$$

[2]I exclude individual earnings shocks to focus on how saving, work, and skill accumulation respond to benefit rules and their financing. This assumption also makes repeated transition and compensation calculations manageable. Assets and skills are the only household state variables. Earnings risk would give households a precautionary saving motive (Conesa and Krueger, 1999; Nishiyama and Smetters, 2007). It could also change their ability to absorb benefit cuts. Thus, the quantitative valuations depend on this assumption.

and the estate x they leave at death:

$$\mathcal{W}(x) = \vartheta \, \frac{(1 + x/b_*)^{1-\sigma} - 1}{1 - \sigma}.$$

The bequest term $\mathcal{W}$ represents a warm-glow motive, as in De Nardi (2004). Households receive utility from the estate they leave. The scale b_* is set in the calibration. Bequests are a luxury good in the model. Households below a wealth threshold leave no estate. Most estates come from wealthy households, as in the data. In year $t+1$, the value of a bequest x is $\mathcal{W}_{t+1}(x) = A_{t+1}^{1-\sigma}\mathcal{W}(x/A_{t+1})$. This adjustment makes the bequest motive consistent with balanced growth.

Skills increase through learning by doing, as in Imai and Keane (2004) and Chang et al. (2002):

$$h_{j+1,t+1} = (1 - \delta_h)h_{j,t} + \eta \, \ell_{j,t} h_{j,t}^{\gamma} \tag{2}$$

where the parameter δ_h is the rate of skill depreciation. Work produces current earnings. It also builds skills that increase future earning capacity. The parameter η measures the efficiency of learning by doing. The parameter γ determines how the skill gain depends on existing skills. The marginal effect of work time on next year's skill is $\partial h_{j+1,t+1}/\partial \ell_{j,t} = \eta h_{j,t}^{\gamma}$. Younger workers have more years before retirement to earn a return on this additional skill. Learning by doing links current work incentives to future earnings. A household can work more to prepare for a benefit cut, building experience as well as current income. Higher benefits and the taxes used to finance them can reduce hours, skills, and future earnings. The model captures these responses through the hours choice.

For a household of age j in year t with assets $a_{j,t}$ and skill $h_{j,t}$, the budget constraint is

$$\begin{aligned} \underbrace{(1+\tau_t^c)c_{j,t}}_{\text{consumption}} + \underbrace{x_{j,t}}_{\text{saving}} + \underbrace{o_{j,t}}_{\substack{\text{medical}\\ \text{bills}}} = {} & \underbrace{[1 + (1-\tau^k)r_t](a_{j,t} + e_{j,t})}_{\text{assets and inheritance, after tax}} \\ & + \underbrace{(1 - \tau_t^{ss} - \tau^G) w_t h_{j,t} \ell_{j,t}}_{\text{after-tax earnings}} \\ & + \underbrace{b_{j,t} + s_{j,t}}_{\substack{\text{Social Security}\\ \text{and other transfers}}} + \underbrace{f_{j,t}(a_{j,t})}_{\substack{\text{floor}\\ \text{payment}}}, \end{aligned} \tag{3}$$

where $o_{j,t}$ is the out-of-pocket share of age-specific medical spending. The term $e_{j,t}$ is the inheritance received. The wage per efficiency unit is w_t. The Social Security benefit is $b_{j,t}$.[3] Other

[3]Benefit amounts in the model are net of the federal income tax on benefits. This tax revenue is credited to the

transfers are $s_{j,t}$, and the consumption-floor payment is $f_{j,t}(a)$.

Assets, skill, and choices differ across households within an age group. Medical bills, inheritances, benefits, and other cash transfers are the same for everyone in an age group in a given year. Floor payments also depend on household assets. The consumption tax rate is τ_t^c. The Social Security payroll tax rate is τ_t^{ss}. The earnings tax rate is τ^G, and the capital-income tax rate is τ^k. The earnings and capital-income tax rates remain at their 2010 values. The consumption tax adjusts when the Treasury must balance its budget. From age 65, the government guarantees consumption of at least $\underline{c}_t$. If a retiree's resources are less than $(1+\tau_t^c)\underline{c}_t$, the retiree receives the difference as the floor payment $f_{j,t}(a)$. The retiree consumes $\underline{c}_t$ and saves nothing. While the retiree receives support, each additional dollar of resources reduces the floor payment by one dollar. This means-tested consumption guarantee follows Hubbard et al. (1995) and De Nardi et al. (2010).

Recursive Formulation A household of age j in year t enters with assets a and skill h and chooses consumption c, saving x, and hours ℓ. With probability $\pi_{j,t}$ it survives to age $j+1$: its saving becomes next year's assets, $a' = \kappa_{j,t}x$, where $\kappa_{j,t}$ is the factor defined in Appendix A.1, and its skill follows (2). With probability $1-\pi_{j,t}$ it dies and its saving is the estate it leaves, valued by $\mathcal{W}_{t+1}$. Its value function satisfies

$$V_{j,t}(a,h) = \max_{c,x,\ell} \Big\{ u(c,\ell) + \beta \big[\pi_{j,t} V_{j+1,t+1}(\kappa_{j,t}x, h_{j+1,t+1}) + (1-\pi_{j,t})\mathcal{W}_{t+1}(x) \big] \Big\}, \tag{4}$$

subject to the budget constraint (3) and the consumption floor, with $\ell = 0$ from age 65. Survival ends at 89, $\pi_{89,t} = 0$, so the last period's saving is left entirely as an estate.

Three features of the solution affect the results. First, the household compares the leisure cost of an additional hour of work with its wage return and skill return. An earnings tax reduces both returns to work. Second, if next year's consumption floor binds, the marginal value of saving is zero. Third, skills and work are complementary. More skill increases the return to work. Work also builds additional skill. With the consumption floor, these effects mean that the lifetime problem need not be concave.

trust fund, so it cancels out of the fund's accounts. Appendix A.2 gives the rate used.

2.3 Estates and firms

Assets left by households that die during the year are collected and distributed the next year as inheritances. This rule follows the distribution of accidental bequests in Huggett (1996) and Braun and Joines (2015). Each age group receives a share ς_j of these estates, with $\sum_j \varsigma_j = 1$. This share is divided equally among the members of that age group. The shares differ across ages and remain fixed over time. They are based on the ages at which people received inheritances in the 2010 Survey of Consumer Finances:

$$\mathcal{E}_{t+1} = \sum_j n_{j,t}(1-\pi_{j,t}) \int x_{j,t}\, d\mu_{j,t}, \qquad e_{j,t} = \varsigma_j \mathcal{E}_t / n_{j,t}, \tag{5}$$

where $\mu_{j,t}$ is the distribution of assets a and skill h among households of age j.

Competitive firms use capital K_t and efficiency units of labor L_t to produce output: $Y_t = K_t^{\alpha}(A_t L_t)^{1-\alpha}$. Factor prices equal marginal products. The wage is $w_t = (1-\alpha)Y_t/L_t$. The net return on capital is $r_t = \alpha Y_t/K_t - \delta$, where δ is the depreciation rate.

2.4 Social Security

The model represents the combined OASDI program with one trust fund and follows the Trustees' projections (Social Security Trustees, 2026). Workers pay a 12.4% payroll tax on covered earnings up to the taxable maximum. The model has neither a taxable maximum nor uncovered work, so it applies the effective rate $\tau_t^{ss} = 0.124\,\zeta_t$ to all earnings, where ζ_t is the projected share of labor income subject to the tax. Payroll receipts are $\mathcal{P}_t = \tau_t^{ss} w_t L_t$.[4]

A retiree's benefit is set in the year the cohort turns 65 as a fixed fraction of average earnings in that year, and it stays constant in real terms for life. It depends on the cohort's average earnings, not on the retiree's own earnings history. This assumption was to make model tractable. Let $b_{j,t}^R$ be the scheduled retired-worker benefit per member of age group j in year t, and $b_{j,t}^O$ the other OASDI cash benefits per member: disability, spouse, child, and survivor benefits, which follow their projected share of scheduled cost. Scheduled spending is

$$\mathcal{H}_t = \sum_j n_{j,t}\big(b_{j,t}^R + b_{j,t}^O\big). \tag{6}$$

Appendix A.5 gives the benefit formula, the adjustment for the rise in the normal retirement age,

[4] Calligraphic letters denote program aggregates and lowercase letters amounts per person.

coverage rates, the income tax on benefits, the benefits already in payment in 2010, and the paths of the taxable share and other benefits.

The trust fund holds reserves $\mathcal{F}_t$ in Treasury securities with a special gross return R_t^F. It receives payroll receipts and Treasury transfers $\mathcal{J}_t$, pays other program expenses $\mathcal{O}_t$, and cannot borrow. It pays only the fraction ϕ_t of scheduled benefits that its resources cover, the paid share:

$$\phi_t = \min\left\{1, \frac{R_t^F \mathcal{F}_t + \mathcal{P}_t + \mathcal{J}_t - \mathcal{O}_t}{\mathcal{H}_t}\right\}, \tag{7}$$

$$\mathcal{F}_{t+1} = R_t^F \mathcal{F}_t + \mathcal{P}_t + \mathcal{J}_t - \phi_t \mathcal{H}_t - \mathcal{O}_t \ \geq 0. \tag{8}$$

Each member of age group j receives $b_{j,t} = \chi_{j,t}\, \phi_t (b_{j,t}^R + b_{j,t}^O)$, where $\chi_{j,t} \geq 1$ is a supplement factor by birth year that equals one unless a reform sets it.

Policy instruments A policy is a path for the payroll rate τ_t^{ss}, a rule for the Treasury transfers $\mathcal{J}_t$, and, for a reform, two cohort-specific instruments. In the baseline the payroll rate stays at 12.4% and the Treasury transfers whatever the fund needs to pay scheduled benefits, as long as it can borrow within its debt limit. Once the limit binds, the transfers shrink to the statutory reimbursements, the fund pays only what its own resources cover, and the paid share falls below one, $\phi_t < 1$. I call these funding-linked benefit cuts. A reform lowers the payroll rate along an announced path and may protect chosen birth cohorts with two instruments: a payroll relief $\upsilon_{j,t} \geq 0$, so that a member of age group j pays the rate $\tau_t^{ss} - \upsilon_{j,t}$ on its earnings, and a benefit supplement $\chi_{j,t} > 1$, which raises the benefit the member receives. The Treasury pays for both: the revenue the relief gives up and the cost of the supplement are added to $\mathcal{J}_t$. Section 4 describes how the two instruments are set in more detail.

2.5 Government spending and the Treasury

Government spending outside Social Security, Z_t, has four parts. Purchases are federal consumption and investment, defense and nondefense, net of depreciation. Cash transfers, the $s_{j,t}$ in the household budget, are the federal transfers other than Social Security and Medicare: unemployment insurance, workers' compensation, nutrition assistance, refundable tax credits, and the remaining federal transfer programs. Medicare is supplied in kind as a share $\mathcal{M}_t$ of output that follows the projected path (Attanasio et al., 2011; Braun and Joines, 2015). Consumption-floor payments are the fourth part. Purchases are a fixed 2010 real amount per person of any age, and each transfer a fixed 2010 amount per person in its recipient group, adults 65 and over or adults

of working age. Spending outside Social Security are imposed exogenously. Households pay the out-of-pocket part of medical spending themselves, the age-specific bills $o_{j,t}$ in the budget constraint.

The Treasury collects income-tax revenue

$$T_t = \tau^G w_t L_t + \tau^k r_t (K_t + B_t).$$

It also collects consumption-tax revenue $\tau_t^c C_t$. After Social Security is combined with the rest of the government, debt evolves as

$$B_{t+1} = (1 + r_t)B_t + \underbrace{\phi_t \mathcal{H}_t + \mathcal{O}_t - \mathcal{P}_t}_{\text{Social Security deficit}} + \underbrace{Z_t - T_t - \tau_t^c C_t}_{\substack{\text{non-Social-Security} \\ \text{primary deficit}}}. \tag{9}$$

The Social Security deficit equals benefits plus other program costs, less payroll receipts. Treasury transfers to the fund cancel on consolidation. The primary deficit of the rest of the government equals purchases, transfers, Medicare, and consumption-floor payments, less income-tax and consumption-tax revenue.

The consumption tax limits next year's debt to $\bar{b}Y_{t+1}$. If debt remain within the limit without this tax, the tax rate is zero. Otherwise, the tax raises the revenue needed to respect the ceiling:

$$\tau_t^c = \max\left\{0, \frac{\widetilde{B}_{t+1} - \bar{b}Y_{t+1}}{C_t}\right\}, \tag{10}$$

where $\widetilde{B}_{t+1}$ is next year's debt with a zero consumption tax. The limit is $\bar{b} = 1$, or 100% of GDP, in the calibration, and $\bar{b} = 1.5$ in the policy experiments. It is an upper bound, not a requirement to hold debt at the ceiling. The government can accumulate assets when revenue exceeds spending and the consumption tax is zero. Lower benefits can therefore lower taxes, repay debt, or build government assets. This fiscal rule follows a debt target in the spirit of Bohn (1998) where the consumption tax enforces the target. Kitao (2014) studies this tax as one way to restore solvency.

In the model, the government uses a consumption tax to finance budget shortfalls when further borrowing would exceed the debt limit. These shortfalls can arise from aging, Medicare spending, or Treasury transfers to Social Security. This tax is a standard choice in models of aging and government debt (Altig et al., 2001; Braun and Joines, 2015; Hansen and İmrohoroğlu, 2016; İmrohoroğlu et al., 2016). It distorts workers' incentives to work, but retirees also pay the tax.

Thus, retirees bear part of the cost of their own benefits.[5]

2.6 Equilibrium

The equilibrium concept is a competitive equilibrium with perfect foresight along a transition path, as in Auerbach and Kotlikoff (1987), Ríos-Rull (1996), and Huggett (1996). Households and firms take prices and policy rules as given, and the price and aggregate paths they expect are the ones their choices produce. Tax revenue is whatever the resulting earnings, capital income, and consumption yield, and the government's budget and financing rules hold in every year.

Definition 1 (Perfect-foresight Competitive Equilibrium). *Let $t_0 = 2010$. Take as given preferences, technology, the demographic paths $\{n_{j,t}, \pi_{j,t}\}$, productivity growth g, the benefit, coverage, spending, Medicare, and consumption-floor rules, the income-tax rates (τ^G, τ^k), the payroll-tax path τ_t^{ss}, and the debt ceiling $\bar{b}$. Initial conditions are K_{t_0}, B_{t_0}, $\mathcal{F}_{t_0}$, benefits already in payment, $\mathcal{E}_{t_0} = 0$, and distributions $\{\mu_{j,t_0}\}_{j=20}^{89}$ with $\sum_j n_{j,t_0} \int a \, d\mu_{j,t_0} = K_{t_0} + B_{t_0}$.*

A perfect-foresight competitive equilibrium is a sequence for $t \geq t_0$ of household policies and values $\{c_{j,t}, x_{j,t}, \ell_{j,t}, V_{j,t}\}$, distributions $\{\mu_{j,t}\}$, prices $\{w_t, r_t\}$, aggregates $\{K_t, L_t, Y_t, I_t, C_t, \mathcal{E}_t\}$, and fiscal outcomes $\{\tau_t^c, \phi_t, B_t, \mathcal{F}_t, Z_t\}$ such that:

1. Households optimize. *$V_{j,t}$ and the policies solve* (4) *subject to* (2), (3), *the consumption floor, $x \geq 0$, and $\ell_{j,t} = 0$ for $j \geq 65$, at the equilibrium prices, taxes, transfers, and inheritances.*

2. Distributions are consistent with household choices. *For $20 \leq j < 89$ and any set of states $\mathcal{D}$,*

$$\mu_{j+1,t+1}(\mathcal{D}) = \int \mathbf{1}\{\mathbf{T}_{j,t}(\mathbf{s}) \in \mathcal{D}\} \, d\mu_{j,t}(\mathbf{s}),$$

   where $\mathbf{T}_{j,t}(\mathbf{s}) = (a', h')$ is the state that a household with state $\mathbf{s} = (a, h)$ reaches at age $j+1$ in year $t+1$, with h' given by (2) *and $\mu_{20,t}$ places entrants at zero assets with the initial skill distribution.*

3. Firms optimize and aggregates are consistent. $Y_t = K_t^{\alpha}(A_t L_t)^{1-\alpha}$, $w_t = (1-\alpha)Y_t/L_t$, $r_t = \alpha Y_t/K_t - \delta$, *and*

$$L_t = \sum_{j=20}^{64} n_{j,t} \int h\ell_{j,t} \, d\mu_{j,t}, \quad C_t = \sum_{j=20}^{89} n_{j,t} \int c_{j,t} \, d\mu_{j,t},$$

[5] Nishiyama and Smetters (2005) show that replacing a progressive income tax with a flat consumption tax can reduce efficiency when households cannot insure against wage shocks. That channel is absent here. The model has no individual earnings risk. It also adds the consumption tax to existing income taxes instead of replacing them.

with estates given by (5).

4. Government rules hold. *The trust fund follows* (7)–(8)*, Treasury transfers $\mathcal{J}_t$ follow the policy's rule, the Treasury satisfies* (9)*, and the consumption tax satisfies* (10)*, equivalently*

$$\tau_t^c \geq 0, \qquad B_{t+1} \leq \bar{b} Y_{t+1}, \qquad \tau_t^c \big(\bar{b} Y_{t+1} - B_{t+1}\big) = 0. \tag{11}$$

5. Markets clear.

$$K_t + B_t = \sum_{j=20}^{89} n_{j,t} \int a \, d\mu_{j,t} + \mathcal{E}_t, \qquad K_{t+1} + B_{t+1} = \sum_{j=20}^{89} n_{j,t} \int x_{j,t} \, d\mu_{j,t}, \tag{12}$$

and, with $I_t = K_{t+1} - (1-\delta)K_t$,

$$Y_t = C_t + I_t + G_t, \tag{13}$$

where G_t is all other spending on goods and services: government purchases, Medicare, out-of-pocket medical spending, and Social Security administration.

Three remarks. First, Social Security reserves are claims between government accounts. Thus, they do not appear in (12). Treasury reimbursements to the fund cancel when the government accounts are consolidated. Second, a cash transfer gives the household money to spend or save. The transfer itself does not use goods or services. Consumption financed by the transfer is already included in C_t. Adding cash transfers to (13) would double-count this consumption. The goods-market equation follows from combining the household and government budgets with asset-market clearing. Third, the equilibrium determines when Social Security reserves are exhausted, when the debt ceiling binds, and the long-run paid share of scheduled benefits.

Definition 2 (Balanced-growth equilibrium)**.** *A balanced-growth equilibrium is a perfect-foresight competitive equilibrium with stationary demographic inputs. Within each age group, the distribution of $(a/A_t, h)$ is constant. Hours, skills, the interest rate, tax rates, and ϕ_t are also constant. Output, capital, consumption, wages, benefits, and fiscal stocks grow at rate g.*

The transition equilibrium is a fixed point in the space of aggregate paths. I fix a horizon of 750 years after 2010 and take the balanced-growth equilibrium of Definition 2 as the terminal state. A guess consists of the paths of capital K_t, effective labor L_t, estates $\mathcal{E}_t$, consumption C_t, and consumption-floor over the horizon. Given the guess, prices follow from the firm's conditions,

and the fiscal rules deliver the paid share, the consumption tax, debt, and reserves year by year, the consumption tax being computed on the guessed C_t. The household problem is then solved backward from the terminal value functions and the distributions forward from the 2010 initial conditions, and aggregating the resulting choices gives implied paths of the same five objects. An equilibrium is a guess that reproduces itself. A policy announced in 2026 is an unanticipated change: the path through 2025 is the baseline's, every household alive in 2026 starts from its baseline state in that year, and the fixed point is computed from 2026 onward.

3 Calibration

I calibrate the model to the U.S. economy in 2010. Household parameters are chosen in the balanced-growth equilibrium of Definition 2 with 2010 mortality rates and fiscal rules, which I call the 2010 stationary equilibrium. Quantities are detrended by productivity growth, and the trust fund has no reserves in it.

The *baseline* is the transition that determines the state inherited by fiscal reforms in 2026. It starts from the 2010 population, public debt, and trust-fund reserves, together with the calibrated capital stock and household distribution. Projected demographics and program rules then determine the path to 2026 and beyond. Appendix A gives the sources and construction of the calibration inputs.

3.1 Externally assigned parameters

Table A.1 lists the parameters set outside the model and their sources. The capital share is $\alpha = 1/3$, which matches a labor share of two thirds in the national accounts. Productivity grows at $g = 1.13\%$ a year, the real-wage growth assumption of the Trustees' intermediate projection, so that the model and the projection it is compared with share the growth rate that drives payroll and benefits. Depreciation is 6.61% a year. Given the capital share, this is the rate at which a capital–output ratio of 3, the household calibration target, delivers a pre-tax return of 4.5% and an after-tax return near 3%, in line with observed returns on wealth. Utility curvature is $\sigma = 2$, an intertemporal elasticity of substitution of one half. Earnings and health are deterministic, so σ governs how willing households are to shift consumption across years rather than attitudes to risk, and the value is in the range life-cycle models of retirement. The bequest scale b_* is the average retired-worker benefit in payment in 2010. Setting the threshold at a retiree's annual benefit makes estates a luxury good at the scale of De Nardi (2004), and the strength of the motive

Table 1: Jointly calibrated household parameters

Parameter		Value	Principal target	Source	Moment	
					Data	Model
Discount factor	$\widetilde{\beta}$	0.9986	Capital–output ratio	Literature	3.00	3.00
Leisure weight	v	3.820	Hours per day, out of 16	ATUS	3.93	3.93
Learning coefficient	η	0.4507	Wage ratio, ages 40–44 / 20–24	CPS-ORG	1.905	1.905
Skill depreciation	δ_h	0.0471	Wage ratio, ages 60–64 / 40–44	CPS-ORG	1.007	1.007
Bequest strength	ϑ	150.3	Estates / GDP (%)	SCF, SSA	5.38	5.38

Notes: The five parameters are fitted jointly. Rows associate each parameter with its principal economic role. Literature: De Nardi et al. (1999) and Cagetti and De Nardi (2006). ATUS: mean work hours per calendar day in the 2010 American Time Use Survey, for all adults aged 20–64. The mean includes nonworkers and zero-work days. The pooled mean is the hours target. CPS-ORG: 2010 Current Population Survey outgoing rotation groups, workers in the indicated age groups. SCF, SSA: Survey of Consumer Finances wealth and Social Security mortality data. The model estate moment uses the 2010 stationary equilibrium's age profiles, weighted by the 2010 population and scaled to initial private wealth. Appendix A gives sample definitions and moment construction.

is fitted below. The skill exponent $\gamma = 0.05$ is taken from Fan et al. (2024). It makes the skill gain from an hour of work nearly proportional to hours, and it is held fixed when learning efficiency and skill depreciation are fitted, because the two age–wage ratios in the calibration cannot identify a third skill parameter. The capital-income tax rate is 30%, the effective rate in Kitao (2014) that combines corporate and personal taxes on capital income. Appendix A.2 gives the construction of each value.

3.2 Joint calibration to 2010 moments

Five household parameters are fitted jointly to five 2010 moments: the discount factor to the capital–output ratio, the leisure weight to average hours, learning efficiency and skill depreciation to two age–wage ratios, and bequest strength to the flow of estates relative to GDP. At each trial parameter vector I solve the household problem and the stationary distribution, so every moment is an equilibrium outcome. Table 1 reports the fitted values. Each parameter is listed with the moment it mainly governs, although all five are determined together.

The capital–output ratio of 3 pins down patience. The primitive discount factor is $\beta = 1.0099$, which after the adjustment for productivity growth becomes $\widetilde{\beta} = \beta(1+g)^{1-\sigma} = 0.9986$. This is the factor that governs private saving. It is distinct from the social discount factor β^* that weights future cohorts in the policy comparisons.

The hours target sets the level of labor supply. Adults aged 20–64 in the 2010 American Time Use Survey work 3.93 hours per calendar day on average, counting nonworkers and days without

work. The leisure weight matches this pooled mean in the 2010 stationary equilibrium, and hours at each age are model outcomes.

The estate target sets the strength of the bequest motive. I combine SCF wealth by age with SSA death probabilities, scale wealth to capital plus public debt, and obtain an annual flow of estates of 5.38% of GDP. The model counterpart uses the stationary equilibrium's age profiles with the same 2010 population weights and the same scaling.

Human capital and the age–wage profile

The model has no exogenous age-productivity schedule. Wages rise and flatten over the working life only because work builds skill under (2) and depreciation erodes it, so the observed age–wage profile is what disciplines the skill technology. In proportional terms,

$$\frac{h_{j+1} - h_j}{h_j} = \eta \ell_j h_j^{\gamma - 1} - \delta_h, \tag{14}$$

so the learning efficiency η scales the gain from an hour of work and the depreciation rate δ_h the annual loss, with the percentage gain falling as skill accumulates. Average skill at entry is normalized to one.

Two wage ratios from the 2010 CPS outgoing rotation groups identify the two parameters: the ratio of mean hourly wages at ages 40–44 to ages 20–24, 1.905, which captures the rise, and the ratio at ages 60–64 to ages 40–44, 1.007, which captures the plateau.[6] Model wage means apply the same positive-hours restriction as the CPS sample and use the stationary population weights, so that hours affect wages only through skill and through who works. Taking ratios removes the wage per unit of skill, allowing me to identify the parameters. The fitted values are $\eta = 0.4507$ and $\delta_h = 0.0471$.

Figure 1 (Appendix Table A.2) compares all nine age groups, each profile divided by its mean at ages 20–24. The model matches the two targets, 1.905 and 1.007, understates wage growth at ages 25–39, and slightly overstates wages at ages 45–59. The largest gap is 8.5% at ages 25–29. The right panel shows why the profile has its shape. Learning exceeds depreciation early in life. As skill accumulates and hours fall, the gain shrinks relative to the loss, the two cross near age 50, where mean skill peaks, and depreciation dominates thereafter. Households that stop working leave the wage sample but stay in mean skill, which moderates the measured wage decline.

[6]The targets are cross-sectional means. The calibration does not estimate separate cohort effects or target wage dispersion at older ages. Skill paths are deterministic conditional on initial skill and household choices.

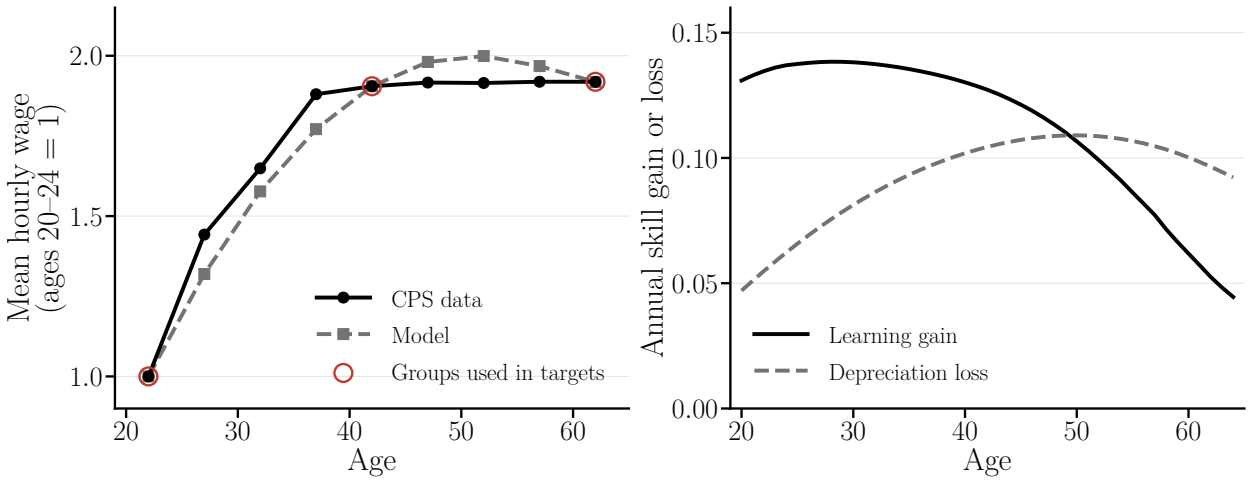


Figure 1: Human capital calibration and the age–wage profile

Notes. Left: mean hourly wages in the 2010 CPS outgoing rotation groups (earnings weights) and the 2010 stationary equilibrium (population weights, households with positive hours), each divided by its own mean at ages 20–24. Points are five-year age groups at their midpoints; red circles mark the two calibration targets. Right: mean skill gained through work and lost through depreciation at each age, over all households; the difference is the change in mean skill. Mean entrant skill is one.

3.3 Initial conditions in 2010

The baseline starts in 2010 from observed aggregates and a constructed cross-section of households. Population and survival probabilities follow the Trustees' projections, with population measured in units of 187 million, the number of adults aged 20–64 in 2010. The old-age dependency ratio rises from 20.7 in 2010 to 45.3 in the terminal population reached in 2190, which is the demographic pressure the baseline carries. Medicare and out-of-pocket medical spending by age come from the National Health Expenditure Accounts, and future Medicare spending follows the 2025 Medicare Trustees projection. Initial capital is three times GDP, the calibration target, public debt and trust-fund reserves take their observed 2010 values, and undistributed estates start at zero. Table 2 reports these stocks together with their 2026 values.

Household heterogeneity

Households share preferences and differ in resources. New cohorts enter at age 20 with zero assets and skills drawn from a lognormal distribution around the mean, with a log standard deviation of 0.493, the dispersion of CPS wages among workers aged 20–24. Skill differences then produce differences in wages, hours, and saving, and wealth and skill evolve jointly through household choices. The cohorts already alive in 2010 receive the same skill dispersion around the 2010 stationary equilibrium's mean skill at their age, and in addition differ in wealth. Assets and skills

Table 2: Initial stocks and the inherited reform state

	2010	2026
Capital / GDP	3.00	3.07
Public debt / GDP (%)	52.98	66.44
Trust-fund reserves / GDP (%)	17.23	8.74
Undistributed estates / GDP (%)	0.00	6.18

Notes: Stocks are measured at the beginning of the year. Initial capital is set to the calibration target. Initial public debt and trust-fund reserves are observed, and undistributed estates start at zero. Mean assets and skills by age follow the 2010 stationary equilibrium. Assets are scaled so that total household wealth equals capital plus public debt. The 2026 column reports the baseline state inherited by all reforms.

are drawn independently within each age. Appendix A.4 gives the numerical construction.

The SCF sets the shape of the wealth distribution within each five-year age group: a mass at zero matching the observed zero-asset share and a lognormal positive part matching the standard deviation of log positive wealth. Zero-asset shares are 30.0% at ages 20–24 and 5.1% at ages 65–69, and the log standard deviations range from 1.67 to 2.02. The level comes from one scale factor applied to the stationary equilibrium's mean asset profile, chosen so that total assets at the 2010 population weights equal capital plus public debt, 3.530 times GDP. The scaling preserves the age profile and the relative dispersion within each age. Figure 2 compares the resulting age means with SCF wealth, which are not targets, and shows the zero-asset shares and the skill distribution. Zero-asset entry raises the model's zero share at ages 20–24 to 44.4%, against 30.0% in the SCF. The other age groups match.

After 2010, inheritances tie wealth accumulation to the estates of the deceased. Estates are distributed across recipient ages in fixed shares taken from the SCF and equally within each age. The estate-flow target in Table 1 sets the strength of the bequest motive that generates them.

Every reform starts from the same state in 2026: the baseline's distribution of assets and skills by household and its capital, undistributed estates, public debt, and trust-fund reserves, which Table 2 reports next to their 2010 values. Before 2026 households expect the baseline, with scheduled benefits while the fund can pay them, funding-linked cuts after reserve exhaustion, no Treasury guarantee, the statutory payroll rate, and the consumption tax that a 100% debt ceiling requires. A reform is announced unexpectedly in 2026, after which households have perfect foresight about it. Preferences and technology are the same under every policy, so common initial conditions isolate the effects of the benefit rules and their financing.

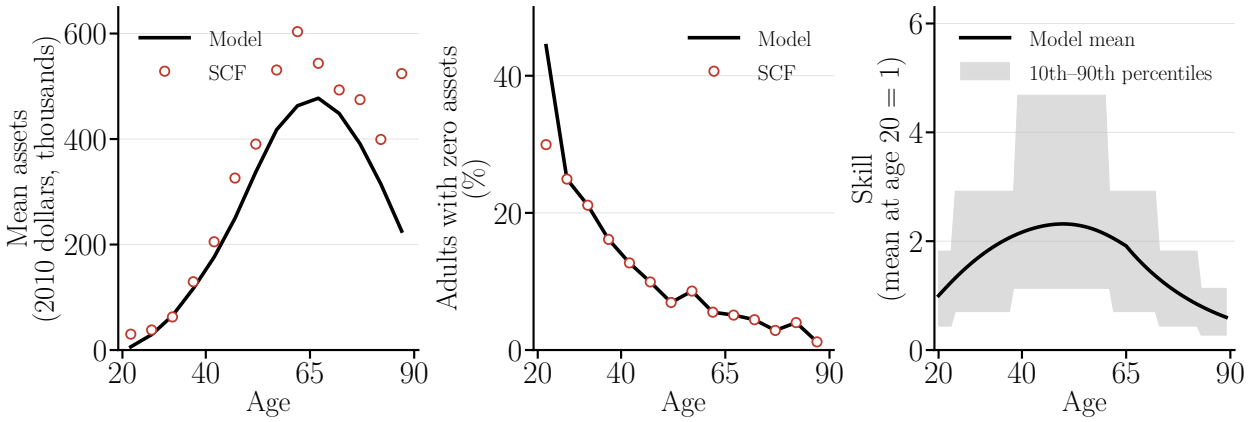


Figure 2: Household heterogeneity at the start of 2010

Notes. Left: mean nonnegative wealth per adult in the 2010 SCF and mean model assets. Center: share of adults with zero assets. Points are five-year age groups at their midpoints; model means use 2010 population weights. SCF family net worth is split equally between spouses, with non-positive values set to zero. Right: mean skill and the 10th–90th percentile range of the initial model distribution, with dispersion from young-worker CPS wages. The band is an assumed cross-section, not a confidence interval or a life-cycle path.

3.4 Fiscal calibration

The 2010 accounts set the levels of the fiscal block and the Trustees' projections set its paths. Appendix A.5 gives every source and construction. Three levels come from the 2010 program accounts. The benefit coefficient, the annualized average benefit of newly entitled 65-year-olds relative to the previous year's taxable earnings per covered worker, is 0.4826, and a fixed factor maps the model's wage bill per working-age adult into taxable earnings per covered worker. Benefits already in payment follow the observed 2010 profile of beneficiaries and benefits by age. Net contributions at the statutory 12.4% rate imply a taxable-payroll share of 0.512, which turns the statutory rate into the effective rate on model earnings. Government purchases and transfers outside Social Security match the 2010 national accounts, with Medicare taken out of cash transfers and supplied in kind so that each expense enters once.

Two rates close the 2010 budget. The earnings-tax rate of 24.6% balances the government budget of the 2010 stationary equilibrium at its debt-to-GDP ratio: spending plus the primary surplus that services the debt net of the borrowing productivity growth allows, less capital-income tax revenue, divided by labor income. Along the transition this rate stays fixed and the consumption tax enforces the debt ceiling. The consumption floor is 9.7% of 2010 GDP per adult, the federal SSI benefit relative to the average retired-worker benefit, and grows with productivity.

The paths are the Trustees' intermediate projections. The statutory payroll rate stays at 12.4%, the taxable-payroll share follows projected payroll receipts relative to labor compensation (Social

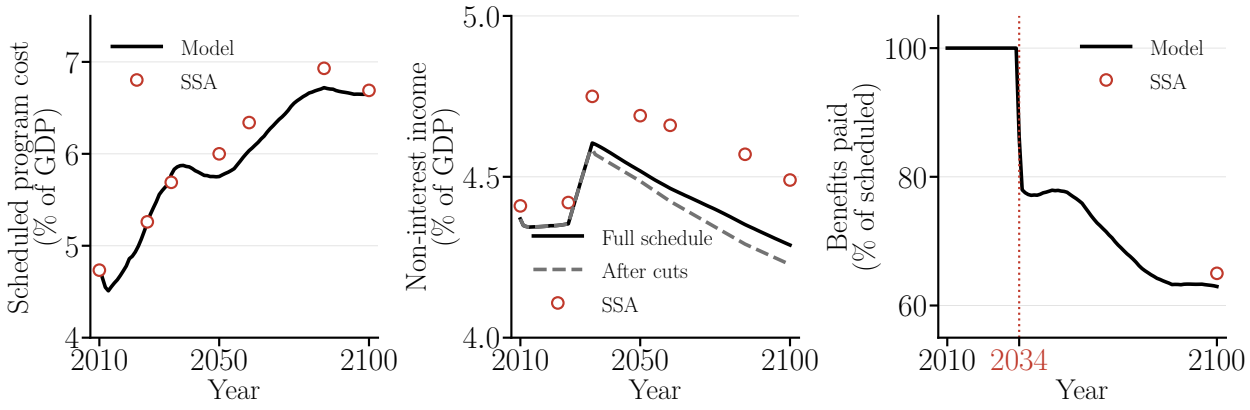


Figure 3: Fiscal calibration: accounts, projections, and model outcomes

Notes. Black lines: baseline. Red circles: 2010 accounts and intermediate projections of the 2026 Trustees Report. Scheduled cost includes gross benefits, administration, and the railroad interchange. Non-interest income includes payroll receipts, taxes on benefits, and specified Treasury reimbursements; the black line assumes full scheduled benefits and the dashed gray line uses benefits paid. The last panel shows the paid share of scheduled benefits, with the Trustees' 2100 projection as the red circle. The vertical line marks reserve exhaustion in 2034. Payroll coverage and benefit composition are projected inputs.

Security Trustees, 2026), and the retired-worker coverage rate and the shares of disability, survivor, and auxiliary benefits follow the decomposition in Appendix A.6. All projected inputs hold their 2100 values thereafter.

Figure 3 compares the baseline with the projections on the three statistics that describe the program's funding pressure: the reserve-exhaustion year, and scheduled cost and non-interest income relative to GDP in 2100. Reserves run out in 2034, the Trustees' year.[7] Scheduled cost reaches 6.65% of GDP in 2100 against 6.69% in the projection. Non-interest income is 4.29% of GDP against 4.49%, and the gap comes mainly from the tax on benefits, which the model applies at a fixed 2.5% while the SSA accounts imply about 5.3% by 2100. The lower income lowers the paid share: the model pays 63.0% of scheduled benefits in 2100 against 65.0% in the projection, and with benefits actually paid, non-interest income is 4.23% of GDP. Because payroll coverage and benefit composition are imposed projections, this comparison measures the fit of the model's demographics, labor supply, and fund accounting conditional on those inputs.

3.5 Model fit outside the targets

The calibration targets five household moments and three fiscal statistics. Appendix Table A.2 reports the model against the data on the moments that were not targeted: wages at the six age

[7]The exhaustion year matches. The within-year reserve balance is not targeted.

groups outside the two ratios, the level of wealth at every age, the zero-asset share of entrants, and Social Security cost and income in 2034 and 2050 together with the paid share in 2100.

The wage profile is matched at its two anchor points and is slightly less concave than the data in between. The model understates wage growth at ages 25–39, by 8.5% at ages 25–29, and overstates wages at ages 45–59 by up to 4%. With the skill exponent fixed, learning efficiency and depreciation can match the overall rise and the plateau but not the speed of the early rise, so the model has young workers accumulating skill a little too slowly and middle-aged workers a little too fast.

Wealth by age is untargeted apart from its aggregate level. The model reproduces mean wealth at ages 30–44 within about 15% of the SCF and the zero-asset shares at every age from 25. It falls short at the two ends. Entrants hold nothing, so mean wealth at ages 20–24 is $6,000 against $30,000 and the zero-asset share is 44.4% against 30.0%. At ages 45–64 the model is 13–23% below the SCF, and at ages 85–89 it holds less than half of the SCF mean. Model households run assets down in old age because mortality is their only risk and estates are a luxury good, while the SCF's oldest respondents are survivors who are selected on wealth and hold assets against medical and longevity risk the model does not have. The model therefore understates the private wealth of the oldest retirees, which makes Social Security a larger share of their resources than in the data and the late-life benefit supplement of Section 4 correspondingly more valuable to them.

The fiscal block is close to the projections in the years it does not target. Scheduled cost is within 0.1% of GDP of the Trustees' figure in 2034 and 0.25% below it in 2050, and non-interest income is 0.15 to 0.2% of GDP below the projection from 2034 on, for the benefit-tax reason given above. The paid share in 2100 is 63.0% against 65.0%. The baseline therefore carries the same funding pressure as the projection, with a shortfall that is marginally larger because of the lower benefit-tax receipts.

4 Reforming Social Security without Cohort Losses

4.1 The Baseline Economy

The baseline is the path the economy follows if Social Security's rules stay as they are: a 12.4% payroll tax, the scheduled benefit formula, and benefits limited to what the program can pay once its reserves are gone. Figure 4 plots it from 2026 to 2200. The driving force is demography. The number of adults aged 65 and over per 100 adults aged 20 to 64 rises from 32 in 2026 to 45 by

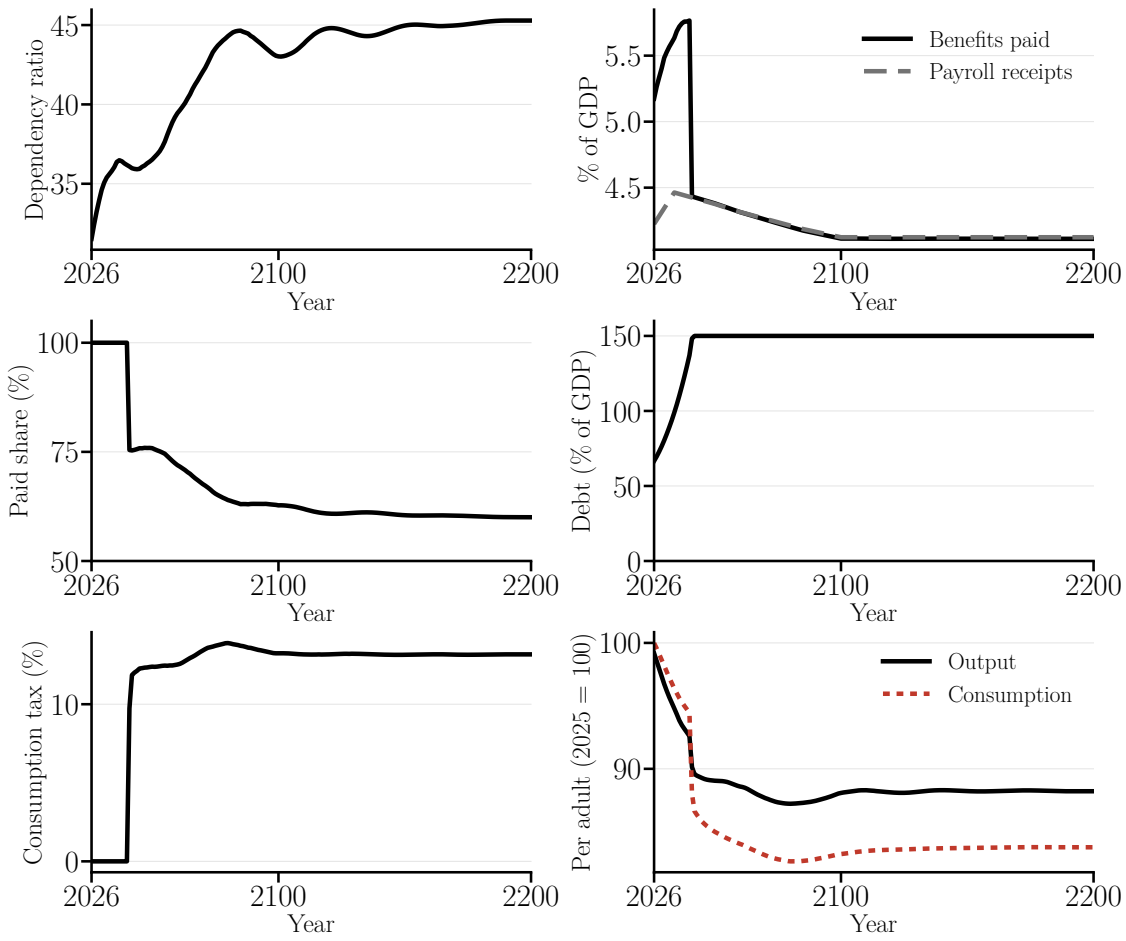


Figure 4: The baseline economy, 2026–2200

Notes. Debt-financed baseline with the 150% ceiling. The old-age dependency ratio is adults aged 65–89 per 100 adults aged 20–64. Benefits paid and payroll receipts are shares of GDP. The paid share is benefits paid over scheduled benefits, on a scale starting at 50. Output and consumption per adult aged 20–89 are indices with 2025 = 100.

2150, where the population becomes stationary. Benefits already exceed payroll receipts in 2026, 5.2% of GDP against 4.2%, and the gap widens to 1.3% of GDP by 2040. Reserves run out in 2034. The Treasury then borrows to pay full benefits, and public debt rises from 67% of GDP in 2026 to the 150% ceiling in 2041. In that year two things happen at once: the funding-linked benefit cuts begin and the paid share of scheduled benefits falls to 75%, what payroll revenue covers, and the consumption tax, zero until then, jumps to 9.7% to hold debt at the ceiling. As aging continues, the paid share drifts down to 72% in 2060 and 60% in 2200, and the consumption tax settles between 12% and 14%.

The economy shrinks per adult without any change in policy. Output per adult falls 10% below its 2025 level by 2041 and 12% by 2080, and hours per adult fall almost as much, for two reasons. The share of adults of working age falls from 76% in 2026 to 69% in 2200, and retirees supply no

hours, so hours per adult fall with that share. This accounts for the decline through 2040. In 2041 the consumption tax starts, and hours per worker drop 4% in that one year. A consumption tax pulls hours in two directions: an hour of work buys fewer goods, which makes leisure cheaper and lowers hours, while lifetime resources buy fewer goods, which makes households poorer and raises hours. For a household taxed in every year of its life, the two effects cancel under the Cobb-Douglas composite of consumption and leisure, whatever the curvature σ, and hours are unchanged.[8]

However, for the cohorts at work in 2041, they do not cancel because much of their lifetime consumption falls before the tax: only the second half of their lives is taxed, and, having seen the tax coming, they bought more goods before 2041 while goods were untaxed. The income effect is therefore weaker than the substitution effect, and these cohorts work less. The size of the drop reflects the low intertemporal elasticity of substitution, one half at $\sigma = 2$: these households cut consumption by less than the tax when it arrives, so their spending in the taxed years rises relative to their wage, whereas a household with an elasticity of one would show no drop at all. As these cohorts retire and are replaced by cohorts that face the tax over their whole working lives, hours per worker return almost to their 2025 level. Output per adult follows hours, with a little extra decline before 2041 because the capital-output ratio decreases from 3.1 to 2.9 while households dissave.

Consumption per adult follows a different path. Until 2040 it holds up better than output, because the Treasury borrows to pay full benefits to a growing number of retirees, who consume out of those benefits while saving falls: investment drops from 20% of output in 2026 to 17% in 2040 and the capital-output ratio from 3.1 to 2.9, so consumption's share of output rises. In 2041 the benefit cut and the consumption tax end this borrowing, saving recovers, and investment returns to 20% of output. At the same time spending on goods outside private consumption, mainly Medicare and out-of-pocket medical care, which rise with age, grows from 17% of output in 2026 to 21% by 2060. Consumption per adult therefore ends 16% below its 2025 level against 12% for output. These declines are the background against which every reform in this paper is measured:

[8]The cancellation needs only the composite form. With utility over $c(1-\ell)^v$ raised to any power, the first-order condition for hours is $v\,c\,(1+\tau^c)/(1-\ell) = (1-\tau^{ss}-\tau^G)\,w_t h$, so leisure is proportional to spending on consumption divided by the after-tax wage, whatever σ. A consumption tax applied in every year of life scales each year's consumption by $1/(1+\tau^c)$ and leaves spending unchanged, so hours are unchanged; this is the same property that keeps hours constant along a balanced-growth path when wages grow. The curvature matters only across the date the tax starts. For a household at work in 2041, the Euler equation across the jump, holding leisure fixed and with $\beta(1+r) \approx 1$, gives $c_{2041}/c_{2040} = (1+\tau^c_{2041})^{-1/\sigma}$. Only with $\sigma = 1$ does consumption fall one for one with the tax, leaving spending and hours unchanged at the jump; with $\sigma = 2$ consumption falls by less, spending rises relative to the wage, and hours fall.

the later figures are relative to 2025, so a reform line that falls is not evidence that the reform fails, and a cohort's welfare change is relative to this path, on which retirees after 2041 already receive a quarter less than scheduled. The 150% ceiling is above the 100% limit that households expected before the announcement. It is what lets the Treasury carry scheduled benefits for seven years after reserves deplete. Under the perfect-foresight assumption, it does not affect their comparison.

4.2 Policy Design and Welfare Measurement

Table 3 lists the three policies. All three are announced in 2026 and face the fiscal rule of the baseline: the Treasury borrows until public debt reaches 150% of GDP, and from then on the consumption tax balances the budget while the earnings and capital-income tax rates stay at their 2010 values.

The main reform keeps the baseline rules through 2060 and then lowers the payroll tax in equal annual steps to 3.5% in 2200. Benefits remain tied to payroll revenue, so the paid share of scheduled benefits falls with the tax, to 16% in 2200. On top of this schedule, two cohort-specific instruments, a payroll rate below the economy-wide rate for some cohorts and a benefit above the paid benefit for others, are financed from general revenue. The uncompensated phase-out, which lowers the payroll tax from 2026 and compensates no one, is the comparison case of Section 5.

I measure welfare by birth cohort. For each household, the consumption equivalent is the uniform percentage change in consumption in every remaining year of life under the baseline that would make the household as well off as under the reform. A cohort's consumption equivalent, CEV below, is the population-weighted average of its members' values across their wealth and skill states. I evaluate 804 cohorts: the 70 born between 1937 and 2006, who are alive in 2026, and the 734 that enter the economy between 2027 and 2760. A reform is Pareto improving in this cohort-average when all 804 are at or above zero.

To rank policy reforms, I aggregate lifetime utilities with the social welfare function

$$W(\beta^*) = \sum_j \omega_j \, V_j + \sum_{t>2026} \left[\beta^* \, (1+g)^{1-\sigma}\right]^{t-2026} \omega_t \, V_t, \tag{15}$$

where V_j is the lifetime utility of a cohort of age j in 2026 and $\omega_j = n_{j,2026} / \sum_i n_{i,2026}$ its share of the adults alive in 2026, V_t is the same object for the cohort entering in year t and ω_t its size at entry relative to the adults alive in 2026, and β^* is the social discount factor on a cohort born one year later, distinct from the household discount factor β. The factor $(1+g)^{1-\sigma}$ removes the effect of productivity growth on utility levels, and cohorts entering after 2510 enter as a constant

Table 3: Payroll-tax and benefit rules in the three scenarios

Policy	Payroll-tax rule	Benefit payments and financing
Baseline	12.4% in every year.	Pay full scheduled benefits through 2040, using Treasury borrowing after reserves run out in 2034. From 2041, limit benefits to program income: 60.02% of scheduled benefits in 2200.
Main reform	12.4% through 2060; then gradually declines to 3.50% in 2200.	Pay full scheduled benefits through 2040. From 2041, pay partial benefits, with further cuts as the payroll tax falls: 16.06% of scheduled benefits in 2200. Treasury-financed payroll-tax relief and benefit supplements protect transition cohorts (Table 4).
Phase-out	Gradually declines from 12.4% in 2026 to zero in 2200.	After reserves run out in 2033, pay benefits only from program income; the Treasury does not borrow to cover the program shortfall. Benefits fall to zero in 2200.

Notes. Scheduled benefits are the promised pensions before funding-linked cuts. All policies are announced in 2026. The debt ceiling of 150% of GDP first binds in 2041 under the baseline and main reform and in 2047 under the phase-out, and consumption taxes close the remaining budget gap. Payroll rates exclude cohort-specific relief.

tail. ΔW is W under the reform minus W under the baseline, in utility units per adult alive in 2026. At $\beta^* = 0$ only those alive in 2026 count; at $\beta^* = 0.97$, the value I highlight, the planner discounts future cohorts at about 3% a year; at $\beta^* = 1$ all cohorts count equally.

4.3 Welfare across Generations

Figure 5 gives the result, with and without the compensation; Table B.1 in the appendix lists the headline numbers. None of the 804 cohorts loses. The smallest gain, 0.00008% of lifetime consumption, belongs to the cohort born in 1937, the oldest alive in 2026, and the gains of those already retired or close to it rise with birth year to 0.07% for the cohort born in 1975. The cut does not touch these cohorts, since it starts in 2060. They gain a little because households that expect lower payroll taxes and lower benefits after 2060 work and save slightly more in the meantime, so output and the consumption-tax base are larger and the consumption tax after 2041 is marginally lower. The cohorts born between 1981 and 2043 are the transition cohorts. Without compensation they would lose. The relief and the supplement are sized to offset those losses and no more, so these cohorts end up just above zero, with gains between 0.01% and 0.06%. Gains become visible for cohorts born after 2045, who spend their working lives under a falling payroll tax: 0.3% for the cohort born in 2049, 0.95% for 2060, 3.4% for 2100, and 7.7% in the long run.

In social welfare terms, ΔW is $+0.04$ utility units per 2026 adult at $\beta^* = 0$, $+0.59$ at $\beta^* = 0.97$, and $+16.6$ at $\beta^* = 1$. The gain is small for a planner who counts only people alive today, because

the reform is built to leave them nearly where they were, and it grows with the weight on future cohorts, who receive almost all of the benefit.

The Pareto property is a statement about cohort averages, and within a cohort households differ in wealth and skill. Table B.2 in the appendix values the reform household by household on the asset and skill grid. Among adults alive in 2026, 4.3% are worse off, all of them in the cohorts born between 1979 and 2002, and none by more than 0.17% of consumption. Of the entrant cohorts examined, only those born in 2030 and 2040 contain losers, 2% and 0.3% of their members, by at most 0.03%; no one born in 2050 or later loses, and the worst-off member of the 2060 cohort gains 0.7%. The losers are low-skill members of the transition cohorts: 27% of the bottom skill tercile of those born 1980–2006 loses against none of the top tercile. The pattern follows from the instruments. Benefits do not depend on a household's own earnings, so the benefit cut takes the same dollars from everyone in a cohort and a larger share of consumption from the low-skilled, while the payroll relief is proportional to earnings and returns more to the high-skilled; the supplement, proportional to the paid benefit, is uniform in dollars and offsets part of the difference. With the cohort-level instruments sized on cohort averages, the transition is Pareto improving on average and leaves a small group of low-earning transition households with losses below a fifth of a percent. Section 6.5 shows that the same two instruments, re-solved on each cohort's worst-off household instead of its average, remove these losses at a higher cost.

4.4 Financing the Compensation

Table 4 describes the two instruments as they are realized in equilibrium, and Appendix B.2 describes how their tables are solved. The payroll relief goes to cohorts born between 2001 and 2042, from 2065 until each cohort retires. The cohort born in 2001, which pays the full 12.4% until age 59 and then retires into reduced benefits, receives the largest relief, 2.29 percentage points. Later cohorts, whose careers overlap more with the falling economy-wide rate, receive less relief: 1.5 points for the cohort born in 2013 and 0.25 for the cohort born in 2037. The benefit supplement goes to cohorts born between 1981 and 2040. Those born before 2000 receive it from retirement, those born later from 2090 or age 75, whichever comes first, and it peaks at 8.6% of the paid benefit for the cohort born in 2020. The two bands overlap because the cohorts born around 2000 to 2040 lose on both margins: they pay the old tax for part of their careers and retire into benefits that are already falling. The relief ends in 2106 with the retirement of the 2042 cohort, the supplement in 2129 with the last eligible cohort.

The Treasury pays for both from general revenue. Its transfers peak at 0.46% of GDP in 2065,

Table 4: Compensation eligibility, timing, amounts, and financing

	Payroll-tax relief	Benefit supplement
A. Eligibility, timing, and size		
Birth cohorts	2001–2042	1981–2040
Paid	From 2065 until retirement	From retirement (born before 2000); from 2090 or age 75, whichever comes first (born 2000 or later)
Largest adjustment	2.29 percentage points off the payroll rate (born 2001)	8.63% of the paid benefit (born 2020)
Last payment	2106	2129
B. Financing of both instruments		
Source	Treasury general revenue, raised by the consumption tax	
Peak Treasury transfer	0.46% of GDP in 2065	
Present value of transfers	2.15% of 2026 GDP	
Consumption tax relative to baseline	At most 0.45 percentage points higher, in 2065; lower from 2072 onward	
Borrowing allowance	None; public debt stays at 150% of GDP	

Notes. Payroll relief lists cohorts with peak relief of at least 0.05 percentage points. Supplements are proportional to the benefit otherwise paid. Financing statistics cover Treasury transfers from 2041; present values use the path's interest rates and output. Tax differences are relative to the baseline.

the year the relief starts, and sum to 2.15% of 2026 GDP in present value. Public debt stays at 150% of GDP in every year from 2041, as in the baseline (Figure B.1 in the appendix), so the consumption tax carries the cost. It exceeds the baseline rate by at most 0.45 percentage points, in 2065, and falls below it from 2072 onward, to 8.2% in 2200 against 13.2%. Three forces lower it. Output and the consumption base are larger. Government spending outside Social Security grows with productivity rather than with output, so it shrinks relative to the larger economy. And the interest rate on the public debt falls as capital accumulates, which at a debt of 150% of GDP saves 0.9% of GDP a year by 2200.

Figure 5 shows what the compensation does. Under the same payroll schedule with neither instrument, 35 cohorts born between 1991 and 2025 lose, by at most 0.14%. The two instruments lift exactly these cohorts to zero and leave the rest of the profile unchanged. Neither instrument is enough on its own. With the relief alone, 24 cohorts born between 1991 and 2014 still lose: the relief starts only in 2065, too late to reach the older cohorts in the band. With the supplement

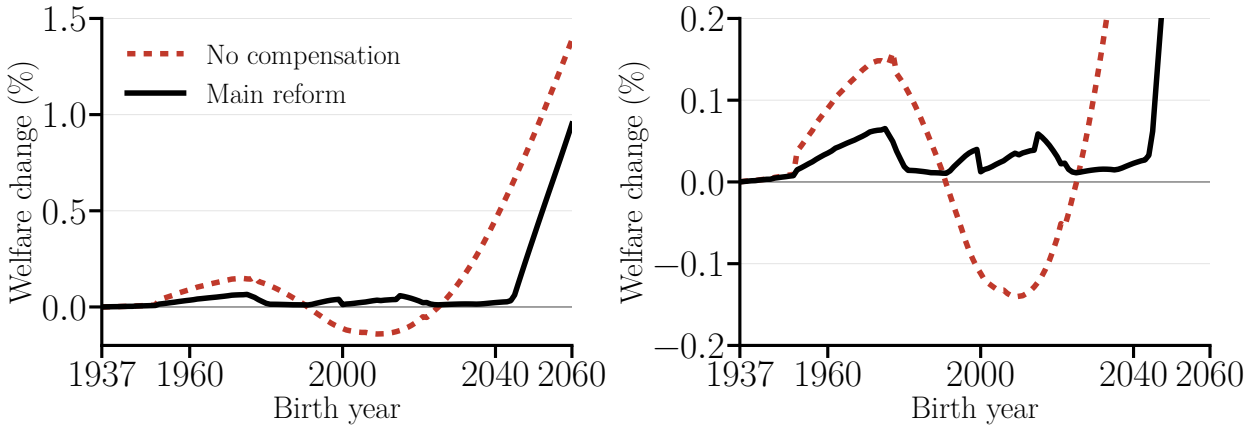


Figure 5: Welfare by birth year under the main reform, with and without the compensation

Notes. Cohort consumption equivalents relative to the debt-financed baseline, births 1937–2060. "No compensation" keeps the main reform's payroll schedule (12.4% to 2060, 3.5% in 2200), dates, Treasury guarantee of the interim paid share, and 150% ceiling but removes both instruments; it is solved in full equilibrium at a market tolerance of 10^{-4}. The right panel narrows the vertical scale.

alone, 30 cohorts born between 2004 and 2033 still lose: a supplement paid late in retirement does not cover the younger cohorts, who also lose from paying the old tax for part of their careers. The cohorts born between 2004 and 2014 lose under either instrument alone, so they need both.

4.5 Macroeconomic Adjustment

Figure 6 plots the adjustment. Until 2060 the two economies are nearly identical. Households learn in 2026 that the payroll tax will fall after 2060 and that their benefits will be smaller, and the two pieces of news push saving in opposite directions: a lower future tax raises expected lifetime income and argues for consuming more now, smaller benefits argue for saving more. The net effect is small, hours per adult 0.2% above the baseline in 2060 and capital per adult 0.05% below, and output per adult differs by 0.2%. Both economies decline relative to 2025 over this period for the demographic reasons.

The reform's effects build as the payroll tax falls, through two channels on the labor side. The lower tax raises the after-tax return to an hour of work, and hours per adult rise 1.6% above the baseline by 2080 and 2.8% by 2100. It also raises the return to the skills that work builds, since each hour now adds to future earnings that are taxed less, and the extra hours themselves produce more learning: skill per hour worked is 1.1% above the baseline in 2100 and 4.2% in 2200. Effective labor, hours times skill, is therefore 4.0% higher in 2100 and 13.4% in 2200, about a third of it from skills.

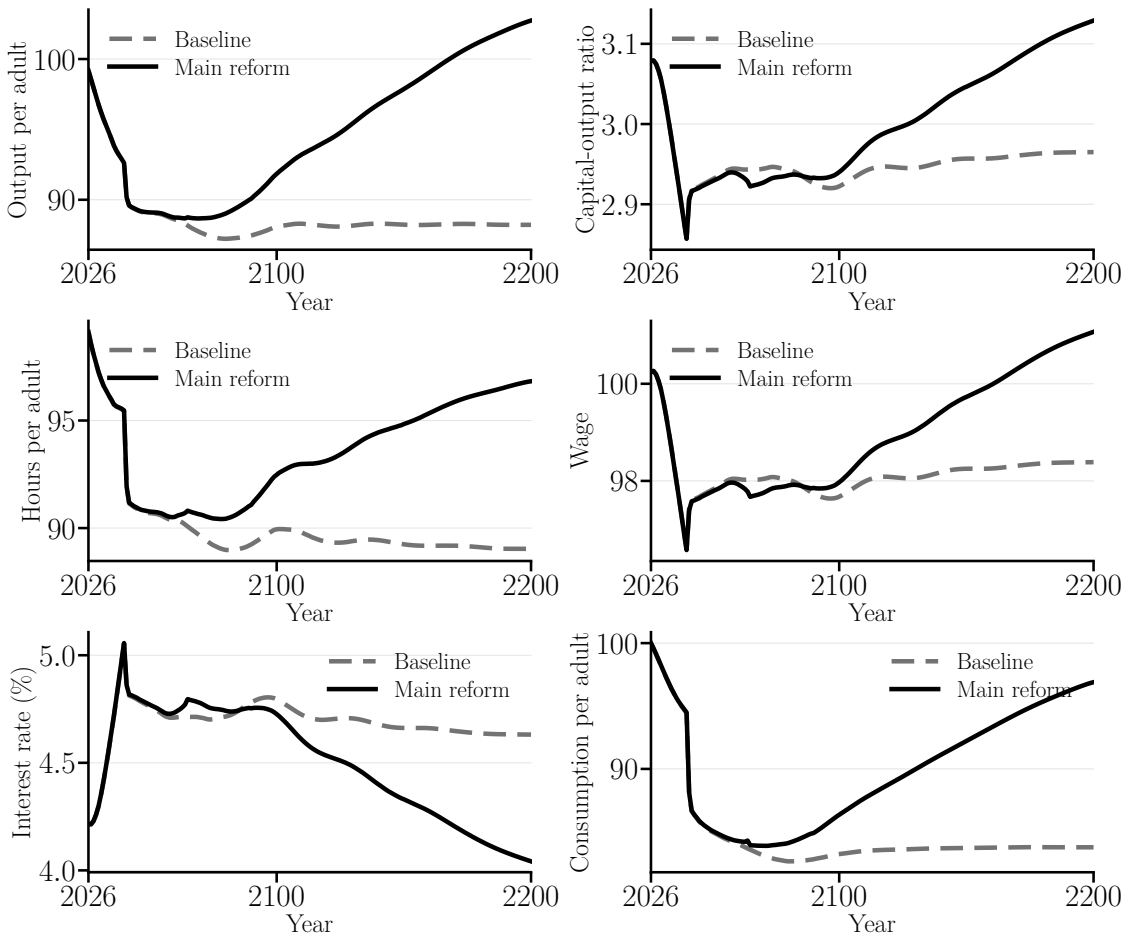


Figure 6: Macroeconomic adjustment under the main reform

Notes. Baseline and main reform, 2026–2200. Output, hours, consumption, and the wage are indices with baseline 2025 = 100, detrended by productivity growth; the first three are per adult aged 20–89, the whole model population. The interest rate is annual, in percent.

On the capital side, households that expect smaller benefits save more for their own retirement. Capital per adult is 4.9% higher in 2100 and 23% higher in 2200, and the capital-output ratio rises from 2.94 to 3.13 while it stays flat in the baseline. The two channels differ in timing. Hours respond as soon as a cohort's own payroll rate falls, whereas capital is a stock that accumulates over the working lives of the cohorts that save more, so for the first decades effective labor grows faster than capital. The capital-labor ratio is below the baseline until about 2090, and the wage per unit of skill is slightly below it too, 0.3% in 2070. Only after capital catches up does the wage rise, by 0.3% in 2100 and 2.7% in 2200. The interest rate falls from 4.6% to 4.0% over the same period. Output per adult is 4.3% higher in 2100 and 16% higher in 2200.

Consumption per adult rises with output, 3.8% above the baseline in 2100 and 16% in 2200, with consumption's share of output unchanged at 59% and the investment share one point higher.

The gain is spread across the two groups of adults, though unevenly. Workers consume 4.5% more than in the baseline in 2100 and 19% more in 2200, as their payroll tax falls and their wage rises. Retirees consume 1.7% more in 2100 and 7% more in 2200 even though benefits fall from 4.1% to 1.1% of GDP, because the retirees of those years saved for their own retirement while working under the lower tax, and because the consumption tax they pay is 8.2% in 2200 instead of 13.2%. The lower interest rate is the one price that moves against savers, and it is part of why the cohorts that hold assets through the transition gain little on their own and need the compensation. These are the resources that pay for the compensation and lower the consumption tax after 2072.

5 Retaining Social Security versus Phasing It Out

I compare the main reform against the uncompensated phase-out policy. It lowers the payroll tax from 12.4% in 2026 linearly to zero in 2200 and compensates no one (Figure C.1). Because payroll revenue falls from the first year, reserves run out in 2033, a year earlier than in the baseline, and the Treasury does not borrow to cover the shortfall. Social Security pays what payroll revenue covers from 2034 onward, so the paid share of scheduled benefits is 77% in 2034, 37% in 2100, and zero in 2200. The rest of the fiscal rule is the baseline's. Debt reaches the 150% ceiling in 2047 rather than 2041, since the Treasury no longer finances benefits between 2034 and 2040, and the consumption tax then balances the budget at a lower rate than in the baseline. The phase-out is the fastest linear removal of the program.

The comparison proceeds in three steps: the adults alive in 2026, then adds the future generations in social welfare function, and then the benefit, tax, and output paths that produce the gains and losses.

5.1 The Adults Alive in 2026

Figure 7 starts with the people who are alive when the policy is announced. For each age in 2026 it plots the change in expected remaining lifetime utility, policy minus baseline, in the units of the social welfare function of equation (15). Averaged over ages with population weights, each curve gives ΔW at $\beta^* = 0$, where only those alive in 2026 count. The weights, the share of adults at each age in 2026, are in Figure C.5.

The main reform's curve is flat and lies just above zero at every age, with an average of $+0.04$. This is because the payroll tax cut does not begin until 2060, and the relief and the supplement hold the transition cohorts at zero, so no one alive in 2026 pays for it.

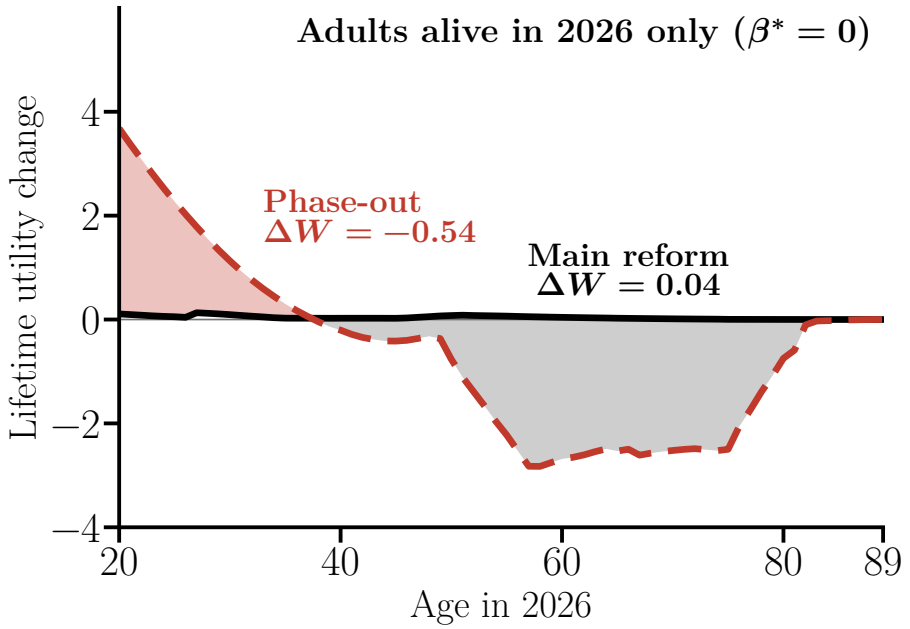


Figure 7: Welfare of the adults alive in 2026 under the main reform and the phase-out

Notes. Change in expected remaining lifetime utility relative to the debt-financed baseline, by age in 2026 (births 1937–2006), in the utility units of equation (15). Each point is a cohort average over wealth and skill states. Averaged with the population weights of Figure C.5 in the appendix, each curve gives ΔW at $\beta^* = 0$. Shading: red where the phase-out lies above the main reform, gray where it lies below; with these weights, red minus gray is the gap between the two ΔW. Figure C.4 in the appendix shows these cohorts and later ones as consumption equivalents by birth year.

The phase-out's curve crosses zero at age 38. Adults younger than that gain, and the youngest gain the most: the 20-year-olds, born in 2006, gain 3.7 units, or 0.9% of their remaining lifetime consumption, because the payroll tax they pay falls from their first working year and keeps falling over their careers. Everyone between 38 and 88 loses. The loss grows with age to between cohorts born between 1949 and 1969, who have paid the full payroll tax for most or all of their careers and whose benefits are cut from 2033. In consumption terms, the loss keeps rising with age, to 2.4% for the cohort born in 1970 and 5.5% for the cohort born in 1951, because the same utility loss spread over fewer remaining years is a larger share of each year's consumption. Above 77 the loss shrinks because fewer years of cut benefits remain.

In sum, 68% of the adults alive in 2026 are worse off under the phase-out, 41% lose more than 1% of their remaining lifetime consumption, and 27% lose more than 3%. The population-weighted average is -0.54 units, a loss of 1.3% of remaining lifetime consumption for the average adult. The loss belongs to the old: adults aged 62 and over contribute -0.59 units and adults under 62 contribute $+0.04$. A planner who counts only the people alive today therefore ranks the main reform above the phase-out.

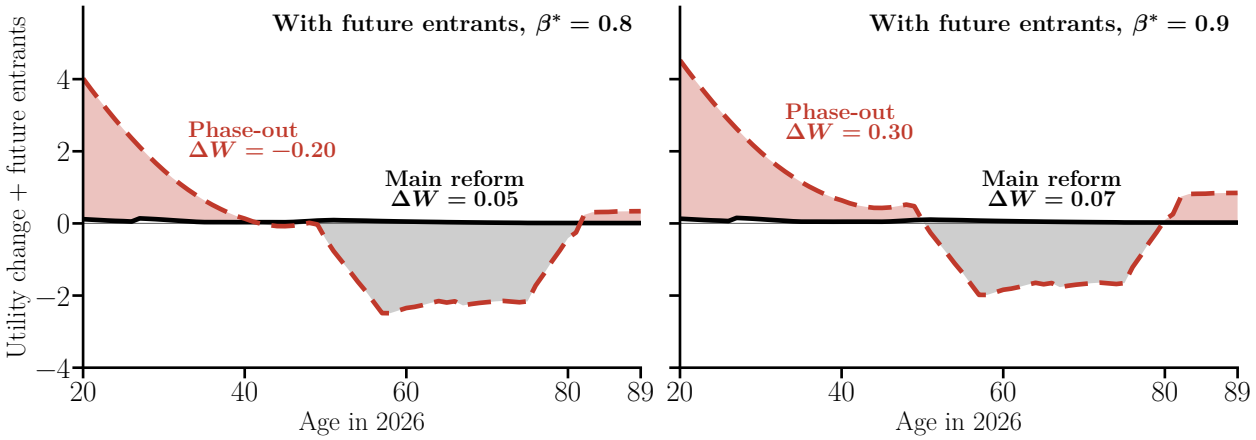


Figure 8: Welfare in 2026 with future generations at two social discount factors

Notes. Each curve adds to its curve in Figure 7 the policy's discounted sum of future entrants' utility changes per 2026 adult, the second term of equation (15), with a constant tail after 2510. Its population-weighted average is therefore ΔW at the panel's β^*. The sum is 0.01 for the main reform and 0.34 for the phase-out at $\beta^* = 0.8$, and 0.02 and 0.84 at $\beta^* = 0.9$. Shading and vertical scale follow Figure 7.

5.2 Adding Future Generations

Next, I add the cohorts that enter after 2026 in the social welfare function, each weighted by adjusted social discount rate. Then ΔW becomes the sum of the average in Figure 7 and a discounted sum of future entrants' gains. Figure 8 shows what the addition does. Each panel plots the curves of Figure 7 shifted up by the policy's discounted sum, so that the population-weighted average of each curve is $\Delta W(\beta^*)$. The left panel uses $\beta^* = 0.8$, a social discount rate of 25% a year on the next cohort, and the right panel $\beta^* = 0.9$, a rate of 11%.

Future entrants gain a great deal from the phase-out and almost nothing from the main reform, so the phase-out's curve rises with β^* while the main reform's barely moves. At $\beta^* = 0.8$ the rise is not enough: more than half of the adults alive in 2026 remain below zero, and the phase-out's ΔW is still negative while the main reform's is positive. At $\beta^* = 0.9$ it is enough. The phase-out's ΔW is now positive and above the main reform's, even though two fifths of those alive in 2026 still lie below zero. The crossing comes at $\beta^* = 0.87$. These adults fare the same in both panels; the panels differ only in the weight the planner gives the people born after them. Table 5 carries the comparison to more patient planners, for whom the phase-out is preferred.

The gains of the two policies start at different times. The phase-out lowers the payroll tax from 2026, so its gains start with the first cohort that enters. The main reform lowers the payroll tax after 2060. Its gains start with the cohorts born after 2045, the first cohorts that work under the lower tax. The social welfare function gives each future cohort a weight that falls with its year

Table 5: Social welfare at selected discount factors

β^*	0.00	0.20	0.40	0.60	0.80	0.90	0.95	0.97	0.98	0.99	1.00
Main reform	+0.04	+0.04	+0.05	+0.05	+0.05	+0.07	+0.18	+0.59	+1.38	+3.99	+16.59
Phase-out	−0.54	−0.52	−0.49	−0.43	−0.20	+0.30	+1.44	+3.09	+5.14	+10.26	+29.94

Notes. Change in utility per 2026 adult relative to the debt-financed baseline. Future-cohort weights are proportional to $[\beta^*(1+g)^{1-\sigma}]^{t-2026}$, with $g = 1.13\%$, $\sigma = 2$, and a constant tail after 2510. The reference discount factor is $\beta^* = 0.97$.

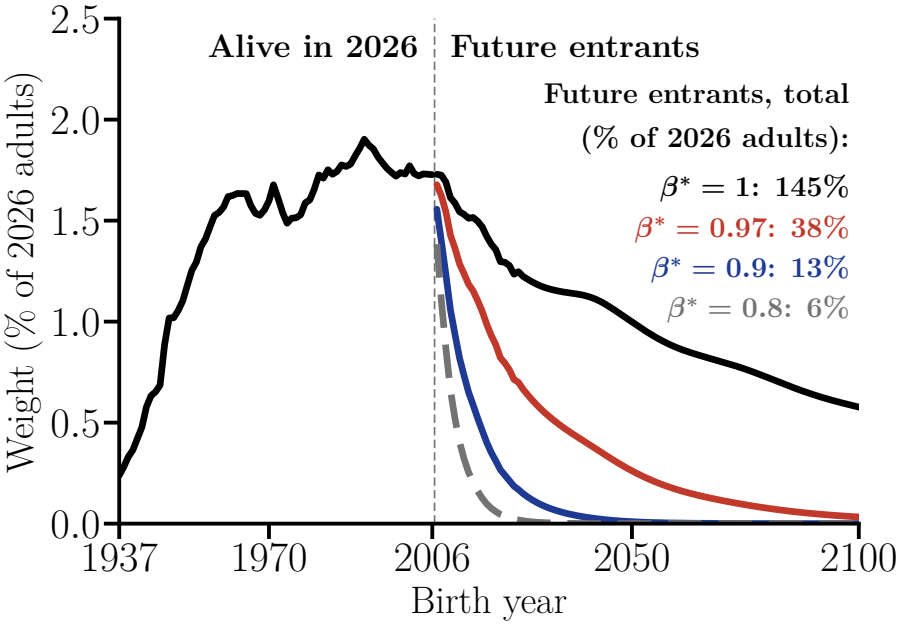


Figure 9: The planner's weight on each generation, by social discount factor

Notes. Weights of equation (15) by birth year, in percent of the adults alive in 2026. A cohort alive in 2026 (born 1937–2006, left of the dashed line) has its population share ω_j, which does not depend on β^*; these shares sum to 100%. A cohort entering in year $t > 2026$ has $[\beta^*(1+g)^{1-\sigma}]^{t-2026}$ times its entry size relative to those adults, with $g = 1.13\%$ and $\sigma = 2$. The key sums the weights of all future entrants, including a constant tail after 2510.

of entry, and Figure 9 plots these weights next to the population shares of the cohorts alive in 2026. The weight falls faster when β^* is lower: all future entrants together carry 6% of the weight of the 2026 adults at $\beta^* = 0.8$, 13% at 0.9, 38% at 0.97, and 145% at $\beta^* = 1$. The first cohorts to enter keep part of their weight at every β^*. The cohorts that enter after 2065 keep almost none of it at $\beta^* = 0.8$ or 0.9, and a large part only when β^* is close to one. The phase-out's gains go to the first cohorts, so some of them count at every β^*. The main reform's gains go to the cohorts that enter after 2065, so they count only for a patient planner. The trade-off is between two gains. The main reform gives a small, certain gain to the people alive today. The phase-out gives a large gain to the people not yet born, and the retirees of today pay for it with their losses. The social discount factor decides which gain counts more.

The comparison has two readings. For every social discount factor up to 0.87, the main reform is preferred to the uncompensated phase-out. For higher discount factors the phase-out's long-run gains dominate its transition losses.

5.3 Benefits, Taxes, and Output

Where do the losses of the adults alive in 2026 and the gains of the cohorts after them come from? Both policies end with a smaller program and a larger economy, so the difference between them lies in the transition. Three things happen at different times under the two policies: benefits are cut, the payroll tax falls, and output rises. Each one affects a different group. The benefit cut reaches the retired, who can do nothing about it. The lower payroll tax reaches the people at work. The extra output reaches the workers who produce it and the retirees they later become. The appendix plots the paths behind this account, the policy rules in Figure C.1, the macroeconomic series in Figure C.2, and the consumption of retirees and workers year by year in Figure C.3.

Benefit cut. The first difference is the benefit cut. For the adults who are retired or close to it, it is the only part of the policy that matters, and their losses in Figure 7 come from it. The phase-out collects less payroll revenue from its first year, so the trust fund runs out in 2033, and the Treasury does not borrow to fill the gap. Benefits fall in 2034, and they keep falling with the payroll rate until they reach zero in 2200. The main reform pays the baseline benefit until 2060. The retirees of 2034 no longer pay the payroll tax, so its fall gives them nothing. They can no longer work more, so the higher return to work gives them nothing either. They respond by drawing down their assets, so their consumption falls by less than their benefits, and the rest of the loss falls on their bequests.

Figure 10 shows why the cut costs these cohorts so much more than the fall in their consumption suggests. Take the cohort born in 1951, aged 75 in 2026, which loses 5.5% of its remaining lifetime consumption. Its consumption, added up over the years it has left, falls by only 2.5%. That average is weighted by dollars, and most of a cohort's consumption belongs to its wealthy members, who barely cut. The welfare measure counts people. Benefits are the same dollar amount for everyone in a cohort, so the cut takes a far larger share of consumption from a retiree with little wealth. Averaged over people, the cohort loses 4.9%. Its bottom wealth tercile loses 6.5%, its top tercile 2.9%, and every household in it loses; the same holds for nearly every household in the cohorts aged 50 to 80. Bequests add the rest. Households value the estate they leave, and the retirees who run down assets to protect their consumption leave smaller estates. For the cohorts aged 65 to 80,

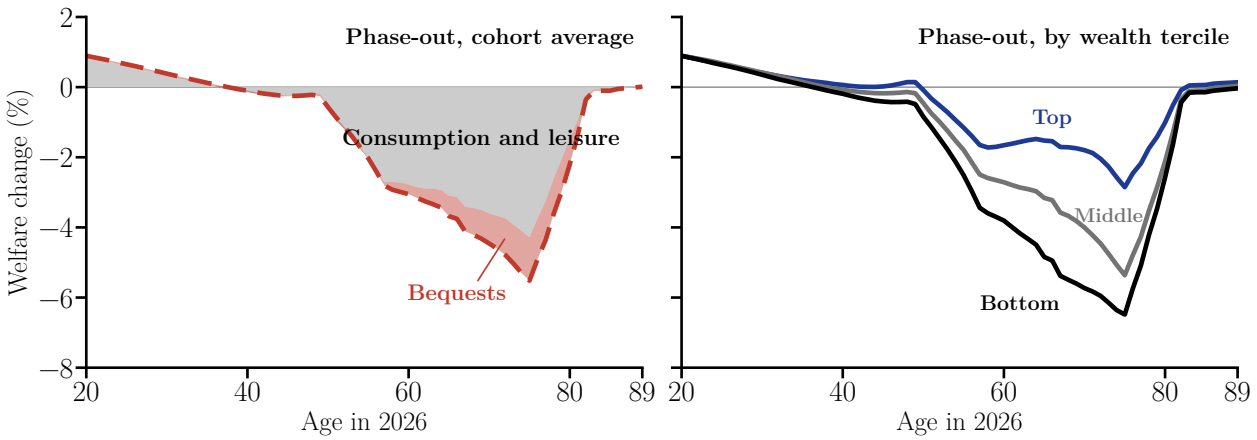


Figure 10: Why the phase-out's losses are large: what the loss is made of and who bears it

Notes. Phase-out against the debt-financed baseline, adults alive in 2026, in percent of remaining lifetime consumption. Left: the cohort-average consumption equivalent, split into a part through consumption and leisure, with the value of bequests at its baseline level, and the remainder through the warm-glow bequest term. Right: household consumption equivalents on the asset and skill grid, averaged within each cohort's wealth terciles. The cohort average values expected utility across households, so in losing cohorts it lies below the person-weighted mean of household values.

the bequest term is between a sixth and a third of the loss, 1.2 of the 5.5 points for the cohort born in 1951. The timing does the rest: these cohorts die before the baseline's own cut in 2041, so they receive none of the offset that later retirees get.

Payroll tax. The second difference is the payroll tax. The phase-out lowers it from 2026 while the main reform holds it at 12.4% until 2060. Workers keep the difference, and their consumption rises above the baseline in the first years of the phase-out, while the main reform leaves it unchanged. The consumption tax is lower under the phase-out as well, because the Treasury never borrows to pay benefits and reaches the debt ceiling six years later than in the baseline. In 2200 the consumption tax is 6.3% against 13.2%. The lower consumption tax is the only channel through which the phase-out helps the adults who are retired by 2034, and it arrives after 2040.

Output. The third difference is output. The phase-out moves the economy at once. Hours rise from 2026, because the payroll tax falls from 2026. Saving rises from 2026, because households that expect no benefits save for their own retirement, and the interest rate falls below the baseline from 2033. The main reform is indistinguishable from the baseline until its payroll tax starts to fall in 2060. For the phase-out, its benefit cut comes in 2034 and its consumption tax changes as debt approaches the ceiling, so hours and consumption decline over several years. The long run is larger too. In 2200 output per adult is 24% above the baseline under the phase-out and 16% under the main reform, and consumption per adult is 23% and 16% above. Figure C.2 has the full set

of series. The extra output goes to the people who produce it. Workers consume more than in the baseline from the first years. The retirees of 2050 and later are different people, cohorts that worked under the lower tax and saved for their own retirement, and their consumption is above the baseline too. From 2050 on both groups gain. This is why every aggregate series is higher under the phase-out than under the main reform, even though the cohorts that are retired when benefits are cut lose.

This is why the phase-out can dominate the main reform in every macroeconomic series and still fail to be Pareto improving. The two comparisons add up different things. An aggregate series adds up the consumption of the people who are alive in the same year, whoever they are. In every year after the 2040s those people are, on average, better off under the phase-out: the workers pay a lower payroll tax, and the retirees are cohorts that spent most of their careers under that lower tax and saved for their own retirement. Figure 7 instead adds up the consumption of the same person over the years of that person's life. For a cohort that is retired when benefits are cut in 2034, those years are the cut and nothing else. The higher output after 2050, the lower consumption tax after 2040, and the higher wage of the long run all arrive after those retirees have died. Nothing in the later aggregates reaches them.

The two views do not contradict each other. The aggregates say that the phase-out makes the pie larger: output per adult in 2200 is 24% above the baseline, against 16% under the main reform. The cohort view says who gets the slices. The extra output is produced by the workers of the 2030s onward and consumed by them and by the future retirees. The cohorts that are retired when benefits are cut produce none of it and receive none of it. A larger pie is necessary for a transition in which no one loses, but it is not sufficient. It becomes sufficient only if part of the gain is moved to the cohorts that would otherwise lose, which is what the main reform's relief and supplement do and the phase-out does not.[9]

An aggregate series ranks the two policies by adding up the consumption of the people alive in each year, and the choice of whose consumption to add depends on a social welfare function. A series that is higher in every year after 2050 counts the people alive in those years, most of whom are not yet born in 2026, and counts the people alive in 2026 only in the years they share with everyone else. That is the planner of Section 5.2 with a social discount factor close to one, and it ranks the phase-out first for the same reason: it gives the gains of future entrants enough weight to outweigh the losses of today's retirees. However, the Pareto criterion asks a different

[9]Section 5.4 shows that the same two instruments make a transition to a payroll tax of zero Pareto improving as well, so the losers of the uncompensated phase-out come from its transition, not from its destination.

question, whether each cohort, followed over its own life, is at least as well off as before. Under the phase-out, it does not.

5.4 The Long-Run Payroll Rate

Section 5.3 traced the phase-out's losers during its transition. This section tests that claim and then asks: once every cohort is protected, how far should the payroll tax fall? The uncompensated phase-out differs from the main reform in two ways, where it ends and how it gets there. It ends at a payroll tax of zero rather than 3.5%, and it gets there by cutting from 2026 with no compensation. To separate the two, I keep the main reform's transition unchanged, and move only the destination to zero. The result is a compensated phase-out. No cohort loses, and the long-run gain is 10.5% of lifetime consumption, against 7.7% under the main reform. The same construction works at every long-run rate between zero and 8%.

Table 6: The same transition at other long-run payroll rates

Long-run rate (%)	Losers	Cohort CEV (%)			ΔW per 2026 adult			
		Born 2060	Born 2100	Long run	$\beta^* = 0$	$\beta^* = 0.89$	$\beta^* = 0.97$	$\beta^* = 1$
0	0	1.46	4.62	10.52	0.053	0.062	0.805	22.269
2	0	1.17	3.94	8.94	0.047	0.060	0.681	19.074
3.5 (main reform)	0	0.95	3.42	7.73	0.044	0.064	0.590	16.585
4.8	0	0.79	2.96	6.68	0.035	0.056	0.514	14.418
6.5	0	0.46	2.30	5.26	0.036	0.061	0.388	11.279
8	0	0.22	1.71	3.99	0.028	0.043	0.269	8.454

Notes. Each row keeps the main reform's dates, financing, debt ceiling, and instruments and changes only the payroll rate reached in 2200, with relief and supplement re-solved until no cohort loses. At 6.5% and 8% the compensation bands widen to births 1937–2060. At 8% the consumption equivalent of the cohort born in 1937 is -8×10^{-7}, below the numerical resolution, and counts as no loss. All rows use the same debt-financed baseline. ΔW is in utility units per 2026 adult, to three decimals because differences near the optimum are thousandths of a unit; 3.5% leads by the widest margin at $\beta^* = 0.89$.

Table 6 reports these compensated transitions. Every row has no losing cohort. The long-run gain falls with the rate, from 10.5% of lifetime consumption at zero to 4.0% at 8%, because a larger program means a higher payroll tax and smaller private saving. Social welfare orders the rates differently at several discount factors. At $\beta^* = 0.97$ and $\beta^* = 1$, where cohorts born after 2035 carry most of the weight, ΔW falls with the rate and zero is best. At $\beta^* = 0$, where only adults alive in 2026 count, the differences across rates are small. In between, at $\beta^* = 0.89$, the 3.5% rate is best. Figure C.6 in the appendix plots ΔW against β^* for each rate: 3.5% leads for discount factors between 0.83 and 0.91, and zero leads outside that range. The question is why a positive payroll tax rate is constrained optimal, when the long-run gain is largest at zero.

Table 7 answers by splitting ΔW at each rate into the contributions of three groups of cohorts. The first is the adults alive in 2026, who are retired or already at work when the reform is announced. The second is the cohorts born between 2007 and 2035, who enter the labor market between 2027 and 2055 and whose working lives straddle the cut that begins in 2060. These are the cohorts the compensation is built for. The third is the cohorts born after 2035, who spend most or all of their working lives under the smaller program. Each group's contribution is the sum of its members' changes in lifetime utility with the social weights, so the three contributions add up to ΔW.

Table 7: Who gains from a larger or smaller program

	0	2	3.5	4.8	6.5	8
Social discount factor $\beta^* = 0.87$						
Alive in 2026, age 62 and over	0.007	0.006	0.004	0.003	0.001	0.001
Alive in 2026, under 62	0.045	0.041	0.040	0.031	0.035	0.027
Born 2007–2035	0.004	0.008	0.015	0.017	0.021	0.013
Born 2036–2100	0.001	0.001	0.001	0.001	0.000	0.000
Born after 2100	0.000	0.000	0.000	0.000	0.000	0.000
Total	0.058	0.056	0.060	0.052	0.057	0.041
Social discount factor $\beta^* = 0.97$						
Alive in 2026, age 62 and over	0.007	0.006	0.004	0.003	0.001	0.001
Alive in 2026, under 62	0.045	0.041	0.040	0.031	0.035	0.027
Born 2007–2035	0.012	0.023	0.035	0.044	0.045	0.031
Born 2036–2100	0.534	0.434	0.355	0.299	0.200	0.129
Born after 2100	0.206	0.178	0.155	0.135	0.107	0.081
Total	0.805	0.681	0.590	0.514	0.388	0.269

Notes. Each entry is a group's contribution to ΔW in utility units per 2026 adult: the population-weighted sum of its cohorts' lifetime-utility changes, with cohorts entering after 2026 discounted by $[\beta^*(1+g)^{1-\sigma}]^{t-2026}$. Columns are long-run payroll rates in percent; 3.5 is the main reform. Rows sum to the total.

Each group responds differently to the long-run rate. The two rows for the adults alive in 2026 barely move as the rate rises: those who need compensation are held at zero, and those who retire before the cut begins in 2060 gain only through prices. The row for the cohorts born 2007 to 2035 rises with the rate up to 6.5%, because at a higher rate fewer of them fall to the floor and more gain on their own. The cohorts that gain on their own at the higher rate are better off there, while those compensated at both rates are at zero either way. The rows for the cohorts born after 2035 fall as the rate rises, because these cohorts spend their careers under the program, and their weight in the sum rises with social discount factor β^*.

The optimal rate depends on how these movements balance. At $\beta^* = 0.87$, moving the

long-run rate from zero to 3.5% lifts the cohorts born 2007 to 2035 by 0.011 units and costs all other cohorts 0.008, so 3.5% wins by 0.002 after rounding. At $\beta^* = 0.97$, the same move costs the cohorts born after 2035 alone 0.23 units, and zero wins.

The key implication is that the interior optimum belongs to the transition, not to the program. In this economy earnings and health are deterministic and inheritances return the assets of the deceased to the living, which leaves little longevity risk to insure, so the long-run welfare of future cohorts rises as the program shrinks, as the Aaron condition predicts when the interest rate exceeds the growth rate. The only reason to keep the payroll tax above zero is the cohorts born 2007–2035. They pay the old tax for part of their careers and retire on benefits that are already falling, so a larger cut makes them lose. Compensation brings a losing cohort back to zero and no further, so it adds nothing to social welfare. With a long-run rate of 3.5% instead of zero, the cut is smaller, fewer of these cohorts lose, and some gain on their own, and those gains do add to social welfare.

Positive optimal sizes in the literature come instead from the insurance that Social Security provides against longevity, earnings, and health risk (İmrohoroğlu et al., 1995; Conesa and Krueger, 1999; Huggett and Ventura, 1999; Hosseini and Shourideh, 2019), and from the program's design. Jang et al. (2025) find that the long-run optimum, measured by the steady-state welfare of a newborn, is zero under the current design and positive once the program is made less distortive and more redistributive. The long-run column of Table 6 is that steady-state measure, and by it the optimum here is zero as well. However, the focus of this paper is the transition. Section 6.2 shows the longevity part of that channel: in an economy without a bequest motive in which households bear their own longevity risk, the uncompensated cut hurts current retirees, the phase-out's worst loss rises to 6.5%, and the range of social discount factors over which retention is preferred widens from 0.80 to 0.84 relative to the fully annuitized economy.

6 Robustness checks

This section asks whether the result depends on learning by doing, on the bequest motive, on how the compensation is financed and how fast the cut proceeds, and on the borrowing allowance, and whether the cohort-average condition can be strengthened so that no household loses. In every variant the two instruments are re-solved cohort by cohort until no cohort loses, and every scenario is scored against the baseline of its own economy. The tables and figures behind each check are in Appendix D.

6.1 Learning by Doing

The channel that distinguishes this model from McGrattan and Prescott (2017) is endogenous human capital where work builds skills. To see how it contributes, I switch it off: skill accumulation and depreciation are set to zero, each cohort's skill by age follows the 2010 age profile of the main economy, so that the age-wage profile is unchanged, and the discount factor and leisure weight are recalibrated to the capital-output and hours targets (Table D.1). The leisure weight falls from 3.82 to 2.40. Without a skill return to work, households must value leisure much less to supply the same hours.

Without the skill channel, the reform's gains are smaller and arrive later. Hours respond about as much as with learning, 8.2% against 8.8% above the baseline in 2200, but the extra hours no longer raise skills, so effective labor rises 6.5% instead of 13.4% and output per adult 10.8% instead of 16.5%. The long-run welfare gain falls by more than output, to 2.8% of lifetime consumption from 7.7%, because workers give up almost as much leisure as with learning but earn less from the extra hours. And the gains arrive a generation later: under the same 3.5% schedule with no compensation, cohort gains turn positive only for births after about 2075, and the worst loss is five times larger.

Smaller and later gains are not enough to pay for the compensation. Re-solving the compensation leaves 107 cohorts worse off. Protecting cohorts born in 1977–2045 shifts the cost to those born in 2046–2092, who have no gains to offset it. The largest loss is 3.1% for the 2046 cohort. Extending compensation through the 2090 birth cohort shifts the cost further into the future. This reverses the mechanism in Section 4.4. With learning, the reform generates gains large and early enough that the consumption tax used to finance compensation falls mainly on cohorts that remain net winners. Without learning, protecting one group of cohorts shifts the cost to the next.

6.2 Bequests and Retirement Resources

The bequest motive affects the result in two ways. It gives households a reason to save even when pension benefits fall. It also generates inheritances that return the wealth of the deceased to the living. These transfers leave households with little longevity risk to insure.

I remove the bequest motive in two economies. The economies differ in how they transfer the assets of the deceased. In both, I set $\vartheta = 0$ to remove the warm-glow term $\mathcal{W}(x)$ from (4). I recalibrate the discount factor, leisure weight, learning efficiency, and skill depreciation to match the same 2010 targets. Demographics, fiscal rules, learning by doing, and the reform remain

unchanged. The discount factor rises from 1.01 to 1.02 to restore the capital-output ratio. The human-capital parameters adjust to preserve the age-wage profile.

In the *annuitized* economy, the assets of deceased cohort members pass to survivors of the same cohort. No inheritances pass between age groups. Private wealth earns a return that depends on survival, as in an actuarially fair annuity. Social Security therefore provides no additional insurance against longevity risk. In the *inheritance* economy, the assets of the deceased return to the living according to recipient age, as in the main economy. Households bear their own longevity risk. Social Security is their only annuity. Appendix Table D.2 compares the three economies. Table D.3 and Figure D.1 in the appendix report the recalibrated parameters, welfare by cohort group, and cohort profiles.

The Pareto result holds without the bequest motive. The main reform leaves no cohort worse off in either economy. The uncompensated phase-out leaves 50 cohorts worse off in the annuitized economy and 51 in the inheritance economy. This pattern is the same as in the main economy (Appendix Table D.2).

The bequest motive changes how much retirees value the annuity that Social Security provides. This effect appears in the losses of cohorts already retired when the uncompensated cut occurs. Under the phase-out, retirees alive in 2026 contribute -0.47 to ΔW when their private wealth is annuitized. Their contribution is -0.59 in the main economy. It is -0.67 when they bear their own longevity risk and Social Security is their only annuity. All values are in utility units per 2026 adult. Without a bequest motive, larger retiree losses increase the range of social discount factors for which retention is preferred. The ranking changes at 0.80 in the annuitized economy and 0.84 in the inheritance economy. In the main economy, which includes the bequest motive, the ranking changes at 0.87.

6.3 Compensation Financing and Implementation Speed

Appendix Table D.4 varies the source of relief funding and the speed of the payroll tax cut. A payroll surcharge on cohorts born in 2044 or later replaces general revenue as the source of relief funding. These cohorts gain most from the reform. No cohort becomes worse off. At $\beta^* = 0.97$, ΔW falls from 0.59 to 0.55. The surcharge reduces the gains of the cohorts that receive the most weight at this discount factor. Delaying completion of the payroll tax cut from 2200 to 2300 also leaves no cohort worse off. At $\beta^* = 0.97$, however, ΔW falls from 0.59 to 0.34. Cohorts born after 2035 pay a higher payroll tax for longer.

Neither a slower phase-out nor a different fiscal rule makes the phase-out Pareto improving.

A phase-out completed in 2300 still leaves 42 cohorts worse off. Under the baseline fiscal rule, the Treasury borrows to pay scheduled benefits until the debt ceiling binds. Applying this rule to the phase-out still leaves 51 cohorts worse off.

6.4 Borrowing Allowances

Appendix Table D.5 reports results with temporary increases in the debt ceiling. Public debt can exceed 150% of GDP while the payroll relief and benefit supplement are paid. Debt must return to 150% of GDP after these payments end. A ceiling of 152% raises ΔW from 0.04 to 0.07 at $\beta^* = 0$. At $\beta^* = 0.97$, it raises ΔW from 0.59 to 0.61. A ceiling of 160% raises ΔW to 0.12 at $\beta^* = 0$ and 0.66 at $\beta^* = 0.97$. Borrowing shifts part of the consumption tax used to finance compensation to later decades, when the tax base is larger. Cohorts alive today therefore gain more. No later cohort becomes worse off. The main results thus hold with a higher debt ceiling.

6.5 Protecting Every Household

The main reform leaves no cohort worse off on average. However, 4.3% of adults alive in 2026 lose who are low-skill members of the transition cohorts. Their losses are at most 0.17% of consumption. I test whether the same two instruments can remove these losses.

I re-solve the compensation using each cohort's lowest consumption-equivalent variation (CEV). This experiment requires CEV to be at least zero at every heterogeneity with positive population weight in all cohorts. Each birth year still has one payroll relief rate and one benefit supplement factor. Neither depends on household assets or skill.

The two instruments can protect every household in the model (Appendix Table D.6). This path uses the same design as the main reform. The macroeconomic paths match to one decimal place, and the long-run gain is unchanged. At $\beta^* = 0.97$, social welfare falls from 0.59 to 0.53 utility units. However, the compensation changes. Each cohort has a uniform supplement factor. Increasing this factor to protect its lowest-skill members therefore increases the supplement for all members. The largest supplement rises from 8.6% to 22% of the paid benefit.

The present value of Treasury transfers rises from 2.2% to 3.3% of 2026 GDP. The consumption tax exceeds the baseline by up to 0.75 percentage points instead of 0.45, and it falls below the baseline from 2091 instead of 2072. The cost rises because the instruments cannot vary with skill: protecting the bottom skill tercile of the transition cohorts also overcompensates the top tercile. Instruments tied to earnings would give the same protection at lower cost, but this exercise shows

that Social Security's own cohort-specific instruments are enough.

7 Conclusion

The trust fund is projected to run out in 2034. Each response to depletion assigns a loss to some group. Following McGrattan and Prescott (2017), this paper studies how to design the transition. I ask whether retaining a smaller program can deliver long-run gains without reducing the average welfare of any birth cohort. Compensation uses Social Security's own instruments without changing income-tax rates.

Such a transition is possible in the model. In the baseline, the Treasury borrows until public debt reaches 150% of GDP. Benefits then fall to the amount that payroll revenue can finance. Relative to this baseline, cutting the payroll tax from 12.4% to 3.5% leaves no cohort worse off on average. Compensation combines a lower payroll rate during working life with a higher benefit late in retirement. I use cohort-specific policy instruments. The general budget finances the compensation. Income-tax rates remain fixed, and the debt ratio remains at its ceiling.

I find that Social Security's own instruments can protect every cohort, even under a complete phase-out. Learning by doing generates the early gains needed to finance compensation. Once cohorts are protected, program size determines how these gains are shared across generations. A smaller program favors distant generations, while a positive payroll rate can preserve more gains for transition cohorts.

The policy debate need not treat scheduled and payable benefits as the only choices. It also need not require a choice between keeping the program and closing it. The gains from a smaller Social Security system can be shared with the transition generations. The task is to design and finance compensation so that welfare gains and losses are shared across generations in a way that minimizes intergenerational conflict. Society can then choose how much of the program to keep through the weights it gives to different generations.

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Appendix

Financing Social Security Reform without Cohort Losses

HongSeok Kim

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A Model and calibration details

This appendix gives the demographic inputs for Section 2 and the parameter, household, and fiscal inputs for Section 3.

A.1 Demographic projections

Figure A.1 shows the population projection and its stationary extension. The figure includes ages 20–99; the model uses ages 20–89.

Net migration. The projected population of an age group can differ from the number of survivors from the previous year. I treat this difference as net migration. In the household problem (4), next year's assets are $a' = \kappa_{j,t} x$ with $\kappa_{j,t} = \pi_{j,t} n_{j,t} / n_{j+1,t+1}$. The numerator is the number of survivors from the current age group and the denominator the size of that group next year, including net migrants. Without migration the two are equal, $\kappa_{j,t} = 1$, and a survivor carries all saving into the next year. With net in-migration, $\kappa_{j,t} < 1$: newcomers bring no assets, so the survivors' saving is pooled across the larger group, which lowers assets per person and preserves the group's total saving.

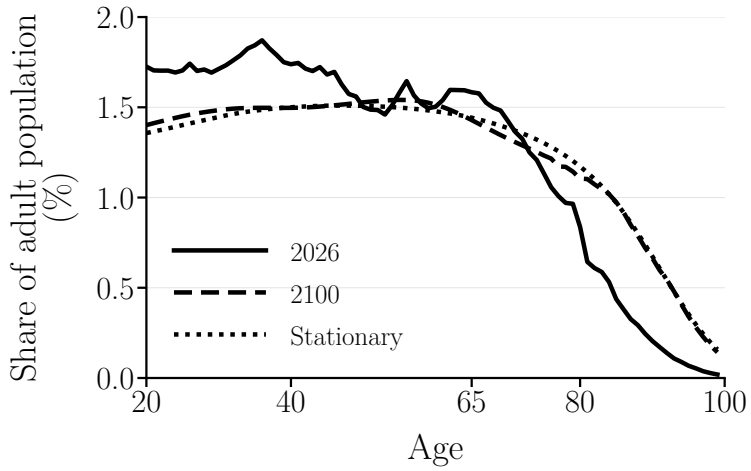


Figure A.1: Projected age distribution of the adult population

Notes. Share of the population aged 20–99 at each age, in percent. The 2026 and 2100 distributions are the intermediate projection of the 2026 Trustees Report. The stationary distribution holds births at their 2100 level and applies the 2099–2100 net survival and migration factors, separately by sex.

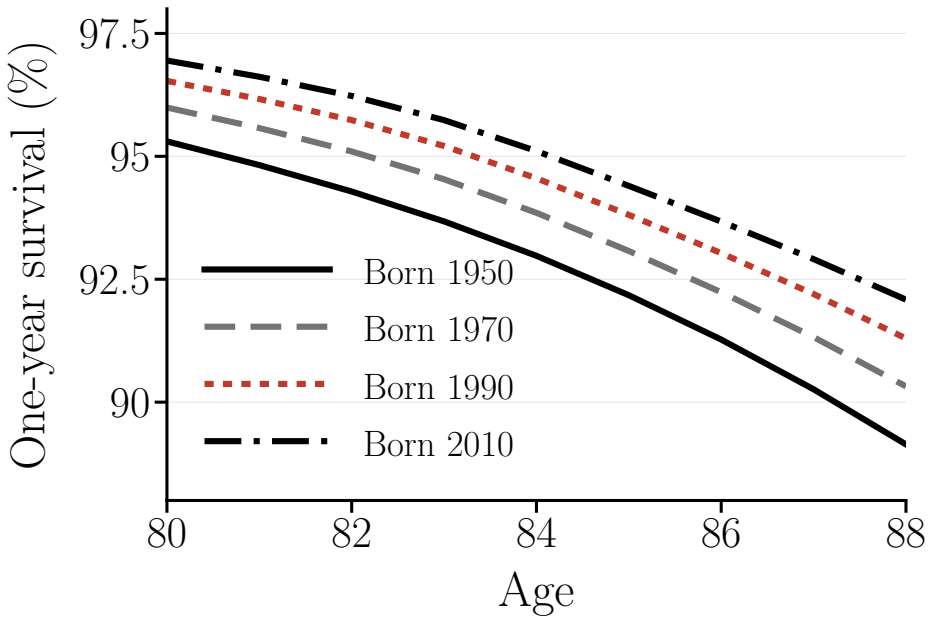


Figure A.2: Annual survival probability by age and birth cohort

Notes. Model survival at ages 80–88 from the intermediate projection of the 2026 Trustees Report, population-weighted across men and women. Cohorts reach the same age in different calendar years. The imposed zero survival at age 89 is omitted.

A.2 Externally assigned parameters

Table A.1 collects the parameters set outside the joint household calibration. Depreciation satisfies $\delta = \alpha/3 - 0.045 = 0.0661$, giving an after-tax return of about 3%.

The bequest scale $b_* = \$15{,}004$ is the count-weighted average retired-worker benefit in payment at the start of 2010. I annualize the 2009 monthly benefits by single age, giving a mean of \$13,946, and multiply by 1.0759 to match 2010 OASDI outlays (Appendix A.5).

The skill exponent $\gamma = 0.05$ is taken from Fan et al. (2024). Their Ben-Porath investment technology differs from (2), so this value is not a direct estimate of learning by doing.

Table A.1: Parameters set outside the model

Parameter	Value	Description and source or basis
Preferences and technology		
σ	2	Utility curvature
b_*	15,004	Bequest scale (2010 dollars). Average retired-worker benefit in payment in 2010, after the scaling to 2010 OASDI outlays (Social Security Administration, 2010; Social Security Trustees, 2011)
g	0.0113	Productivity growth. Real-wage growth in the Trustees' intermediate projection (Social Security Trustees, 2025)
α	$1/3$	Capital share. —
δ	0.0661	Capital depreciation. $K/Y = 3$ (De Nardi et al., 1999; Cagetti and De Nardi, 2006) and a 4.5% pre-tax return
γ	0.05	Skill exponent in learning by doing. Fan et al. (2024)
Social Security		
ρ	0.4826	Retirement benefit at 65 / taxable earnings. 2010 program accounts (Social Security Administration, 2010, 2011)
ζ_{2010}	0.512	Effective taxable-payroll share, using net contributions. Social Security Trustees (2011). Future path from the 2026 Trustees Report
φ_s	0.929	Benefit adjustment for birth cohorts from 1960 onward. Statutory benefit reduction at age 65
τ^b	0.025	Income-tax rate on benefits, credited to the trust fund. Model setting. The 2010 average rate was 3.4%
Taxes, public debt, and the consumption floor		
τ^G	0.246	Earnings-tax rate. Balances the 2010 reference budget
τ^k	0.30	Capital-income tax rate. Kitao (2014)
$\bar{b}$	1.00 / 1.50	Debt ceiling / GDP: calibration and years before 2026 / all policies from 2026 (Section 4.2). CBO projection: debt exceeds 100% of GDP around 2030
$\underline{c}_0$	0.097	Floor / 2010 GDP per adult aged 20–89. 2010 SSI benefit relative to the average retirement benefit (Social Security Administration, 2011)
Health spending		
p_j^M	0.594 / 0.083	Medicare share: ages 65+ / under 65. NHEA tables by age and payer (Centers for Medicare and Medicaid Services, 2012), scaled to 2010 spending
p_j^O	0.14 / 0.22	Out-of-pocket share: ages 65–84 / 85+. NHEA tables by age and payer (Centers for Medicare and Medicaid Services, 2012)
$\mathcal{M}_t$	3.45 / 6.23 / 6.69	Medicare / GDP (%): 2010 / 2050 / 2100. Medicare Trustees (2025)

A.3 Household calibration targets

The five jointly fitted parameters and their targets are in Table 1. This subsection gives the wage and estate calculations behind those targets.

Wages. I use the January–December 2010 CPS outgoing rotation groups. The sample includes workers aged 20–64 with positive hours and earnings. Hourly wages equal weekly earnings divided by usual weekly hours. Mean wages by five-year age group give the two targets: $\$23.20/\$12.18 = 1.905$ for ages 40–44 relative to 20–24, and $\$23.37/\$23.20 = 1.007$ for ages 60–64 relative to 40–44.

Estates. Let n_a be the 2010 population at age a, q_a its SSA death probability, and $\bar{W}_a$ mean SCF wealth per adult. Using the wealth sample in Appendix A.4, the target is

$$\frac{\text{estates}}{\text{GDP}} = \underbrace{\frac{\sum_a n_a q_a \bar{W}_a}{\sum_a n_a \bar{W}_a}}_{\text{share of wealth left by decedents}} \times \underbrace{\frac{K_{2010} + B_{2010}}{Y_{2010}}}_{\text{model wealth / GDP}} = 0.01524 \times 3.530 = 5.378\%.$$

The model counterpart uses the 2010 stationary equilibrium's age profiles with 2010 population weights and the same wealth scaling.

A.4 Initial stocks and household distributions

Units and stocks. With $A_{2010} = 1$, initial effective labor L_0, and $K/Y = 3$, output is $Y_0 = 3^{\alpha/(1-\alpha)} L_0$. The dollar conversion is $D = \$15{,}049 \text{ billion}/Y_0$, using 2010 GDP (Social Security Trustees, 2025). Population is measured relative to adults aged 20–64 in 2010.

Initial capital is $3Y_0$, or $45.147 trillion. Public debt is $0.5298Y_0$, based on $7,811 billion held by the public in December 2009 (Treasury). Trust-fund reserves are $\mathcal{F}_{2010} = 0.1723Y_0$: OASDI assets were $2,592 billion in 2010 dollars (Social Security Trustees, 2011, Table III.A3). Household wealth totals $K_{2010} + B_{2010} = 3.530Y_0$, and the initial undistributed estates is zero.

Wealth and inheritances. The 2010 SCF sample pools respondents and spouses aged 20–89 from 6,482 families. I divide family net worth equally between spouses and set non-positive wealth to zero. I pool the five imputations and divide survey weights by five. Each five-year age group supplies a zero-wealth share and the standard deviation of log positive wealth. These determine a mass at zero and a lognormal positive part. Figure A.3 reports these inputs and the recipient-age shares of inheritances. Each five-year inheritance share is divided equally among its single ages.

Mean initial assets follow the reference age profile, scaled to capital plus public debt:

$$\bar{a}_{j,2010} = \lambda \bar{a}_j^{\text{ref}}, \qquad \lambda = \frac{K_{2010} + B_{2010}}{\sum_{j=20}^{89} n_{j,2010} \bar{a}_j^{\text{ref}}}.$$

This common scaling uses the actual 2010 population and preserves the age profile and relative dispersion. It is separate from the SCF wealth scaling used for the estate target.

Skill dispersion. The standard deviation of log hourly wages among CPS workers aged 20–24 is $s_h = 0.493$, using earnings weights and 13,746 observations. I interpret wage dispersion as skill dispersion and impose a lognormal distribution with this log standard deviation on entrants and all opening cohorts. Mean skill by age follows the 2010 stationary equilibrium.

Asset and skill draws are independent within age because the SCF and CPS samples are not linked. I approximate normal draws with nine Gauss–Legendre nodes per dimension on $[-4, 4]$ and normalize the approximation to preserve means. New age-20 entrants have zero assets.

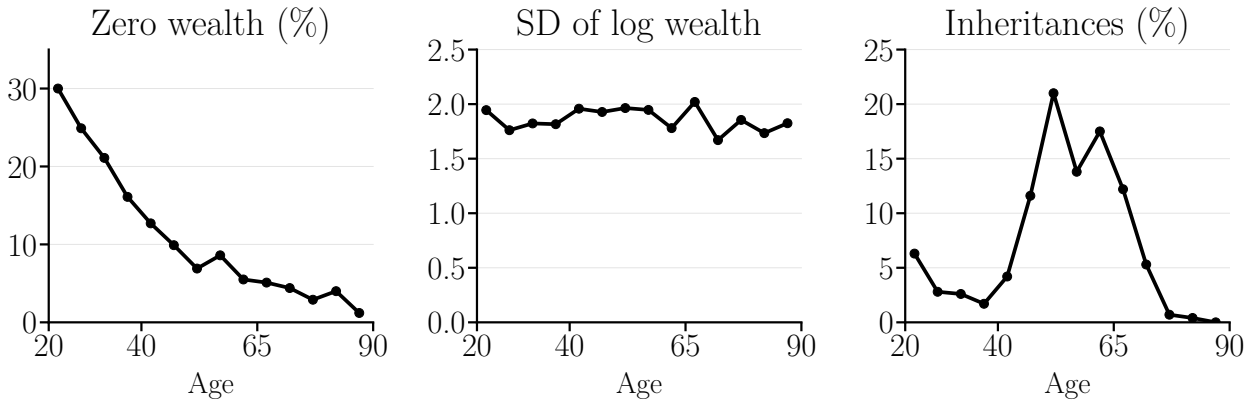


Figure A.3: Distributional inputs from the 2010 SCF

Notes. SCF respondents and spouses aged 20–89, in five-year age groups plotted at their midpoints. Panels show the zero-wealth share, the standard deviation of log positive wealth, and the share of 2010 inheritances by recipient age. Family net worth is split equally between spouses, with non-positive values set to zero.

A.5 Fiscal calibration

Retirement-benefit coefficient. A retiree of age j in year t turned 65 in year $s = t - j + 65$. That cohort's initial age-65 benefit is $\hat{b}_s = \varphi_s \rho \, \bar{e}_{s-1}$, where $\bar{e}_{s-1}$ is taxable earnings per covered worker in the year before the cohort turns 65, ρ is the ratio of the average initial benefit of newly entitled retired workers to these earnings, and φ_s adjusts for the rise in the normal retirement age. The benefit is constant in real terms for life. The coefficient is

$$\rho = \frac{12 \times \$1{,}370}{\$34{,}065} = 0.4826.$$

The numerator annualizes the average benefit of workers newly entitled at age 65 in 2010, at December benefit rates (Social Security Administration, 2011, Table 6.A4). The denominator is 2009 taxable earnings per covered worker in 2010 dollars: \$5,271 billion across 158 million workers gives \$33,375, converted with the CPI ratio $213.967/209.630$ (SSA Annual Statistical Supplement, Table 4.B11).

The model has no taxable maximum or covered-worker count. A fixed factor maps its wage bill per working-age adult into taxable earnings:

$$\bar{e}_t = \psi \frac{w_t L_t}{n_t^W}, \qquad \psi = \frac{\text{2010 taxable payroll per covered worker}}{(1-\alpha) \times \text{2010 GDP per adult aged 20–64}}.$$

Thus, new benefits grow one-for-one with the wage bill per working-age adult. For cohorts born from 1960, $\varphi = 0.867/0.933 = 0.929$ adjusts for the rise in normal retirement age from 66 to 67. Claiming at 65 pays 86.7% rather than 93.3% of the primary insurance amount. I omit the phase-in for births 1955–1959.

Benefits in payment and coverage. Initial benefits use the 2009 retired-worker counts and mean monthly benefits at ages 62–89 (Social Security Administration, 2010, Table 5.A1.1). Grouped counts are split across single ages in proportion to population. A common factor of 1.0759 scales benefit amounts, including disability, survivor, and auxiliary benefits, to 2010 OASDI outlays less lump-sum death payments, \$701 billion (Social Security Trustees, 2011, Table III.A3).

Let ν be the 2010 share of adults aged 65 and over receiving retired-worker benefits. Let ξ_t be beneficiaries per adult aged 65 and over relative to 2010 (Appendix A.6). For $j \geq 65$, the scheduled retired-worker benefit per member is

$$b_{j,t}^R = (1-\tau^b)\,\nu\,\xi_t\,\hat{b}_s, \qquad s = t - j + 65,$$

where $\hat{b}_s$ includes φ_s and τ^b is the benefit-tax rate. Retirees present in 2010 keep their initial benefits. Reform payments are $\chi_{j,t}\phi_t(b_{j,t}^R + b_{j,t}^O)$, as in Section 2.4. The solver carries a multiplier on the retired-worker component of scheduled spending, applied to existing retirees and new beneficiaries alike; it equals one under every policy in this paper and is omitted from the text.

Payroll coverage. The taxable share ζ_t accounts for uncovered work and earnings above the taxable maximum, \$106,800 in 2010 and \$184,500 in 2026. From payroll receipts $\mathcal{P}_t$, I compute

$$\zeta_t = \frac{\tau_t^{ss}}{0.124}, \qquad \tau_t^{ss} = \frac{\mathcal{P}_t}{(1-\alpha)Y_t}.$$

Net contributions of \$637.283 billion imply $\tau_{2010}^{ss} = 0.0635$ and $\zeta_{2010} = 0.512$. The one-time general-fund reimbursement of \$2.4 billion enters separately. The retained SSA comparison

includes the rounded reimbursement in its payroll benchmark, implying $\zeta_{2010} = 0.514$. The model-to-SSA payroll ratio is therefore about 0.9962 throughout the projection.

For 2026–2100, the effective rate is 0.124 times the Trustees' taxable-payroll-to-GDP ratio divided by $1 - \alpha$. I interpolate linearly between 2026, 2034, 2050, 2060, 2085, and 2100 (Social Security Trustees, 2026, Tables IV.B3 and VI.G1), and between the 2010 and 2026 anchors. Coverage remains constant after 2100. The path is imposed relative to projected covered workers, matching projected receipts relative to labor compensation. Under reform, the announced statutory rate replaces 0.124 in $\tau_t^{ss} = \zeta_t \times (\text{statutory rate})_t$. Cohort relief reduces the member's own rate; ζ_t retains its projected path.

Income taxes and spending. The capital-income tax applies to $r(K + B)$. The earnings-tax rate balances the reference budget:

$$\tau^G = \frac{G^{\text{purch}}_{2010} + S^G_{2010} + (r_{2010} - g)B_{2010} - \tau^k r_{2010}(K_{2010} + B_{2010})}{w_{2010}L_{2010}} = 0.246.$$

Purchases are federal consumption and gross investment less fixed-capital consumption and Social Security administration: $1,086 billion, or 7.22% of GDP. Other transfers are federal expenditures less interest, Social Security, and purchases: $1,828 billion, or 12.15% of GDP (Bureau of Economic Analysis, 2011, Tables 3.2 and 3.12). The model replaces observed interest payments with the balanced-growth primary surplus, $(r - g)B$.

Each spending item is a fixed 2010 real amount per person in its group that grows with productivity and with the group's projected size. Purchases are per person of any age, children included. Medicare and other retirement transfers (railroad retirement and pension benefit guaranty) go to adults aged 65 and over. Unemployment insurance, workers' compensation, nutrition assistance, refundable credits, and military medical insurance go to ages 20–64. Remaining transfers are equal per adult. Medicare and military medical insurance are removed from cash transfers and supplied in kind. None of these paths depends on the wage or output of the policy being solved. The debt ceiling is $\bar{b} = 1$ before 2026 and 1.5 in the main policy comparisons, apart from the borrowing variants in Section 6.4. It first binds in 2041 in the baseline and in 2047 under the uncompensated phase-out.

Consumption floor. The floor starts at age 65. It is set from the 2010 federal SSI benefit relative to the average retired-worker benefit, $\$8{,}088/\$13{,}968 = 0.579$ (Social Security Administration, 2011). The implemented floor is 9.7% of 2010 GDP per adult aged 20–89 and grows with productivity. It applies in the calibration and before 2026, then is switched off in the main experiments. Keeping it in the main economy changes no result because it never binds (Section 6.2).

Medical spending. NHEA personal health care spending per person in 2010 is $3,628 at ages 0–18, $4,422 at 19–44, $8,370 at 45–64, $16,872 at 65–84, and $32,411 at 85 and over (Centers for Medicare and Medicaid Services, 2012).

Unscaled Medicare shares are 0.07 below age 65 and 0.50 from age 65. A common factor matches 2010 Medicare outlays of $518 billion, or 3.4% of GDP, giving shares of 0.083 and 0.594. Out-of-pocket shares are 0.11, 0.14, 0.13, 0.14, and 0.22 in the five age groups above.

The Medicare path $\mathcal{M}_t$ sums Part B, Part D, and Hospital Insurance spending relative to GDP. Hospital Insurance spending equals its cost rate times its taxable-payroll-to-GDP ratio. I scale the latter from 0.4355 in 2024 using projected OASDI taxable payroll relative to GDP (Medicare Trustees, 2025, Tables III.B7, III.C10, III.D6).

A.6 Projected fiscal paths

The following inputs complete the payroll-coverage path above. Projected ratios and rates retain their 2100 values thereafter.

- *Retired-worker coverage* ξ_t: beneficiaries per adult aged 65 and over relative to 2010. It is 1 in 2010, 0.997 in 2034, 1.012 in 2050, and 1.044 in 2100 (Social Security Trustees, 2025, Table V.C4).
- *Other benefits* $\mathcal{H}_t^O$: disability, spouse, child, and survivor benefits. They equal 1.5% of GDP in 2010 and retain their projected share of scheduled cost and their 2010 age allocation (Social Security Trustees, 2025, Table IV.B1).
- *Trust-fund return* R_t^F: historical effective yields through 2025 and projected yields from 2026, deflated by the Trustees' CPI (Table VI.G1).
- *Other fund outlays* $\mathcal{O}_t$: administration and vocational rehabilitation, $6.58 billion in 2010, grow with population. The railroad interchange, $4.39 billion, grows with the population aged 65 and over (Social Security Trustees, 2011, Table III.A3).
- *Medicare* $\mathcal{M}_t$: 3.45% of GDP in 2010, 6.23% in 2050, and 6.69% in 2100.

A.7 Calibration fit

Table A.2 reports the moments behind the wage, wealth, and fiscal figures in Section 3. The fiscal comparison retains the income and paid-share gaps discussed there.

Table A.2: Calibration fit: data and model

	Wage ratio (20–24 = 1)		Zero assets (%)		Mean wealth ($1,000)	
Age group	Data	Model	Data	Model	Data	Model
A. Household moments by age group, 2010						
20–24	1.000	1.000	30.0	44.4	30	6
25–29	1.442	1.320	24.9	24.9	38	29
30–34	1.649	1.577	21.1	21.1	63	65
35–39	1.880	1.771	16.1	16.1	129	117
40–44	1.905	1.905	12.7	12.7	205	176
45–49	1.916	1.980	9.9	9.9	326	250
50–54	1.915	1.999	6.9	6.9	390	338
55–59	1.919	1.968	8.6	8.6	531	418
60–64	1.919	1.918	5.5	5.5	604	463
65–69	–	–	5.1	5.1	544	477
70–74	–	–	4.4	4.4	493	449
75–79	–	–	2.9	2.9	475	391
80–84	–	–	4.0	4.0	399	315
85–89	–	–	1.2	1.2	524	225

B. Social Security finances, % of GDP

	Scheduled cost		Non-interest income	
Year	Data	Model	Data	Model
2010	4.73	4.78	4.41	4.37
2034	5.69	5.77	4.75	4.61
2050	6.00	5.75	4.69	4.52
2100	6.69	6.65	4.49	4.29
Reserve exhaustion year	2034	2034		
Paid share in 2100 (%)	65.0	63.0		

Notes. Panel A: wage ratios from the 2010 CPS outgoing rotation groups (earnings weights), relative to ages 20–24, and wealth from the 2010 SCF, with family wealth split equally between spouses. Model wage means use stationary population weights and households with positive hours; model wealth uses the opening 2010 distribution. The fitted targets are the 40–44/20–24 and 60–64/40–44 ratios. Zero-asset shares are imposed from age 25; zero-asset entry changes the 20–24 share. Panel B: baseline against the 2010 accounts and 2026 Trustees projection, with cost and income defined as in Figure 3.

B The main reform

This appendix gives the supporting results and compensation method for Section 4.

B.1 Welfare across and within cohorts

The no-loss condition applies to cohort averages. Tables B.1–B.2 report cohort and household welfare.

Table B.1: Cohort welfare under the main reform

Measure	Main reform
Evaluated birth cohorts	804
Losing cohorts	0
Smallest cohort CEV (%)	$+0.00008$
Birth year of the smallest gain	1937
Cohort born 2060: CEV (%)	$+0.95$
Cohort born 2100: CEV (%)	$+3.42$
Long-run cohort CEV (%)	$+7.73$
Social welfare change ΔW, $\beta^* = 0$	$+0.04$
Social welfare change ΔW, $\beta^* = 0.97$	$+0.59$
Social welfare change ΔW, $\beta^* = 1$	$+16.59$
Peak public debt (% of GDP)	150

Notes. Main reform against the debt-financed baseline at the 150% ceiling. CEV is the cohort-average consumption equivalent; the minimum keeps five decimals to distinguish it from zero. ΔW is in utility units per adult alive in 2026, with a constant tail after 2510.

Table B.2: Within-cohort welfare under the main reform and the household-level path

	Main reform, all households (%)			Losing by skill tercile (%)		Household-level path (%)	
Birth year	Mean	Losing	Worst	Bottom	Top	Mean	Worst
Alive in 2026, by birth year							
Born 1937–1960	+0.016	0.0	−0.001	0.0	0.0	+0.035	+0.001
Born 1961–1979	+0.051	0.0	−0.017	0.0	0.0	+0.019	+0.001
Born 1980–2006	+0.019	9.1	−0.170	27.3	0.0	+0.032	+0.001
All adults alive in 2026	+0.028	4.3	−0.170	13.0	0.0	+0.029	+0.001
Entrant cohorts, by birth year							
Born 2010	+0.033	0.0	+0.018	0.0	0.0	+0.012	+0.001
Born 2020	+0.031	0.0	+0.004	0.0	0.0	+0.029	+0.001
Born 2030	+0.015	2.2	−0.029	6.5	0.0	+0.057	+0.001
Born 2040	+0.023	0.3	−0.018	0.9	0.0	+0.069	+0.001
Born 2050	+0.365	0.0	+0.273	0.0	0.0	+0.085	+0.003
Born 2060	+0.951	0.0	+0.734	0.0	0.0	+0.643	+0.514
Born 2080	+2.243	0.0	+1.711	0.0	0.0	+2.046	+1.566
Born 2130	+5.103	0.0	+3.647	0.0	0.0	+5.093	+3.642
Born 2180	+7.461	0.0	+5.479	0.0	0.0	+7.460	+5.478

Notes. Household CEVs at each asset–skill node relative to the debt-financed baseline. Adults alive in 2026 start from their inherited states, with population weights across groups, and entrants from their opening distributions. Mean is the population-weighted ex-ante CEV used in the text, Losing the share of households with a CEV below -0.01%, Worst the lowest household CEV. Entry-skill terciles each hold one third of cohort mass, splitting boundary-node mass pro rata. The welfare decomposition matches the solver value functions to 10^{-12}. The last two columns value the household-level path of Section 6.5; its losing share is zero in every group.

B.2 Solving for the compensation

The two instruments are tables by birth year b. Payroll relief υ_b, in percentage points, is available to births 2000–2045 from 2065 until retirement. The benefit factor $\chi_b \geq 1$ is available to births 1977–2043. It applies from retirement for cohorts born before 2000, and from 2090 or age 75, whichever comes first, for later cohorts. These are eligibility bands; Table 4 reports the cohorts with realized compensation. Treasury transfers finance both instruments.

Starting from the uncompensated schedule, I solve the full equilibrium with the current tables and update them from each cohort's realized consumption equivalent CEV_b:

$$\upsilon_b \leftarrow \max\{0,\ \upsilon_b + c_\upsilon(\bar{\epsilon} - \mathrm{CEV}_b)\}, \qquad \chi_b \leftarrow \max\{1,\ \chi_b\,[1 + c_\chi(\bar{\epsilon} - \mathrm{CEV}_b)]\}.$$

The target is $\bar{\epsilon} = 0.01\%$ of lifetime consumption, with $c_\upsilon = 2$ and $c_\chi = 12$. Downward steps are damped by one half to limit overshooting below zero. The supplement reaches cohorts already retired when relief begins; relief reaches those still working. The two tables therefore allocate compensation to different parts of the eligibility bands.

Screening rounds use a market-clearing tolerance of 10^{-4}. I then hold the tables fixed and

re-solve at 5×10^{-5}. The main reform is reported as Pareto improving only if all 804 cohort CEVs are non-negative in this verification solve. Robustness cases verified at 10^{-4} are marked in their table notes. For the 6.5% and 8% endpoints in Section 5.4, eligibility expands to births 1937–2060 to reach the affected cohorts; Table 6 states the numerical qualification at 8%.

B.3 Fiscal paths

Figure B.1 shows the debt and consumption-tax paths behind the financing results in Section 4.4.

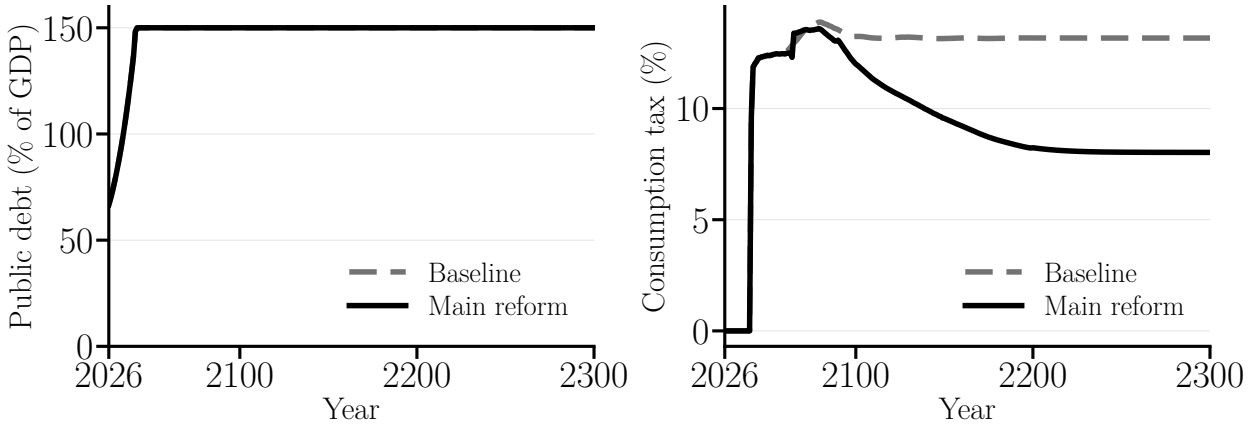


Figure B.1: Public debt and consumption-tax paths

Notes. Baseline and main reform, 2026–2300, both under the 150% ceiling. Debt is measured at the start of the year relative to each policy's GDP. Consumption-tax rates are percentages of consumption expenditure.

C Phase-out and the long-run payroll rate

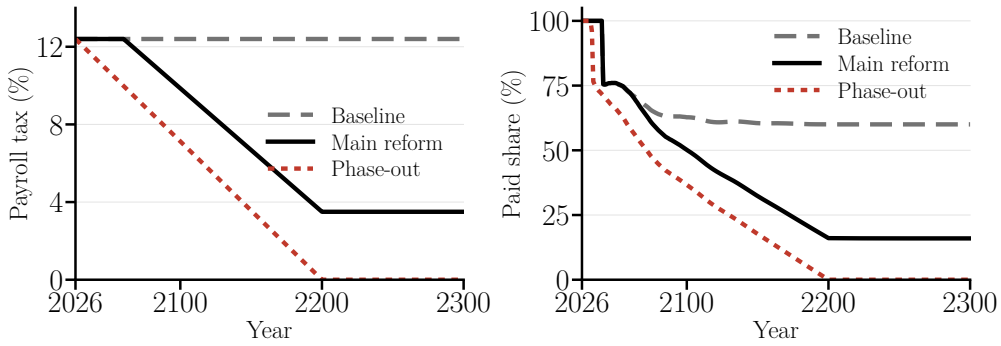


Figure C.1: Payroll-tax and benefit paths across the three scenarios

Notes. Payroll rates and benefits paid as a percentage of scheduled benefits, 2026–2300, all under a 150% debt ceiling. The main-reform payroll line is the economy-wide rate before cohort relief; the others are effective rates.

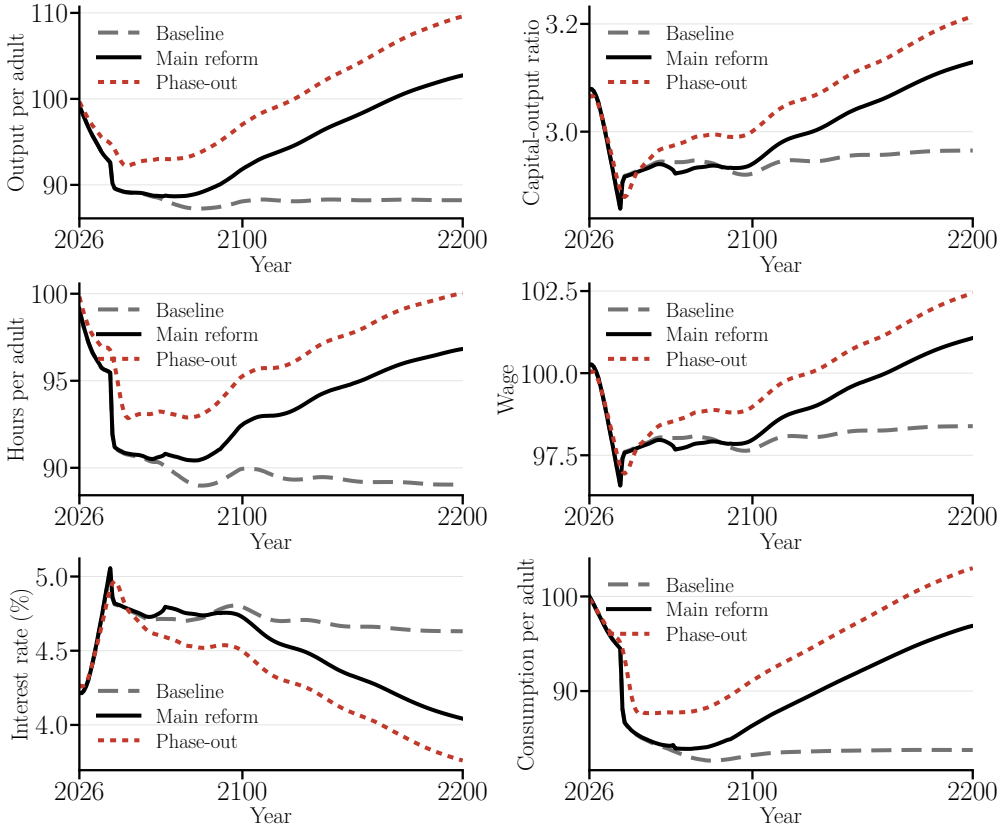


Figure C.2: Macroeconomic adjustment under the main reform and phase-out

Notes. Baseline, main reform, and uncompensated phase-out, 2026–2200. Output, hours, consumption, and wages are indices with baseline 2025 = 100, detrended by productivity growth; the first three are per adult aged 20–89. The interest rate is annual, in percent.

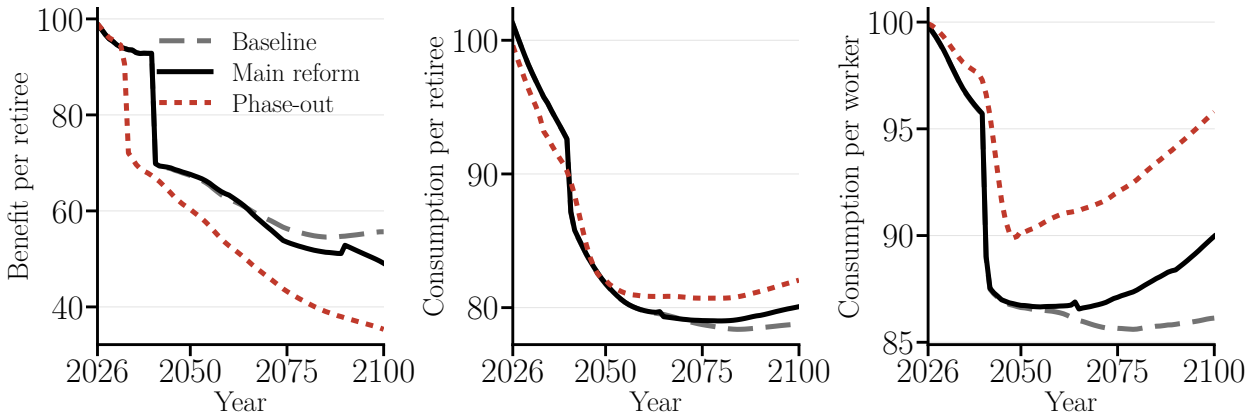


Figure C.3: Benefits and consumption of retirees and workers across the three scenarios

Notes. Indices with baseline 2025 = 100. Retirees are adults aged 65–89 and workers adults aged 20–64. Benefit per retiree is Social Security outlays divided by the number of retirees. Each year's retirees are different cohorts; Figure 7 follows those alive in 2026 over their lives.

C.1 Cohort welfare

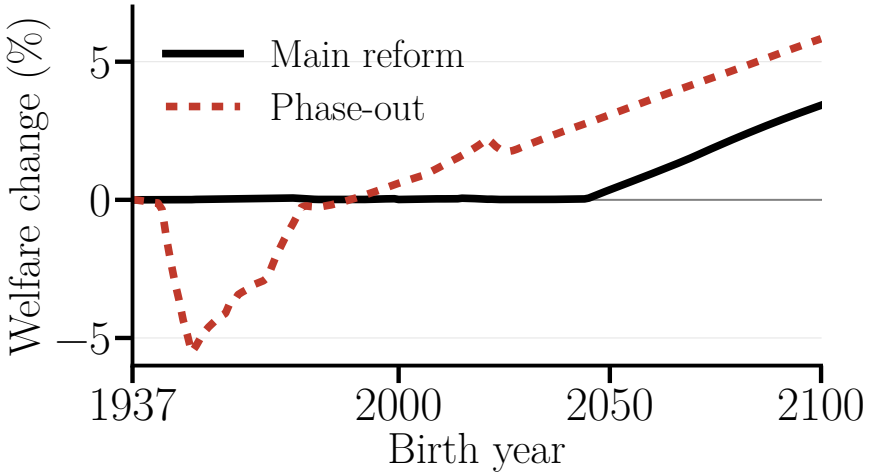


Figure C.4: Cohort welfare under the main reform and phase-out

Notes. Cohort CEVs relative to the common debt-financed baseline, births 1937–2100. Losing-cohort counts use all 804 evaluated cohorts.

C.2 The long-run payroll rate

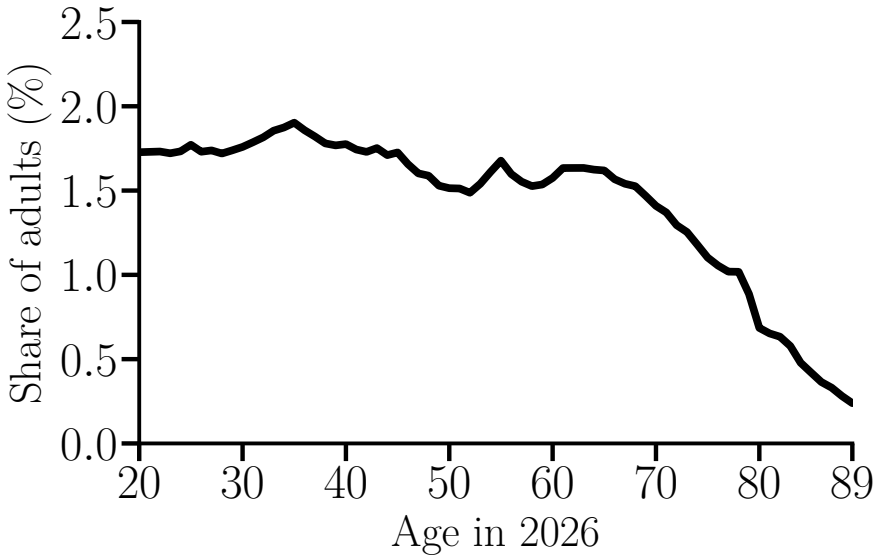


Figure C.5: Population weights by age in 2026

Notes. Share of the model population aged 20–89 at each age in 2026 (births 1937–2006), in percent. These are the weights ω_j of equation (15) and the weights behind the averages in Figures 7 and 8.

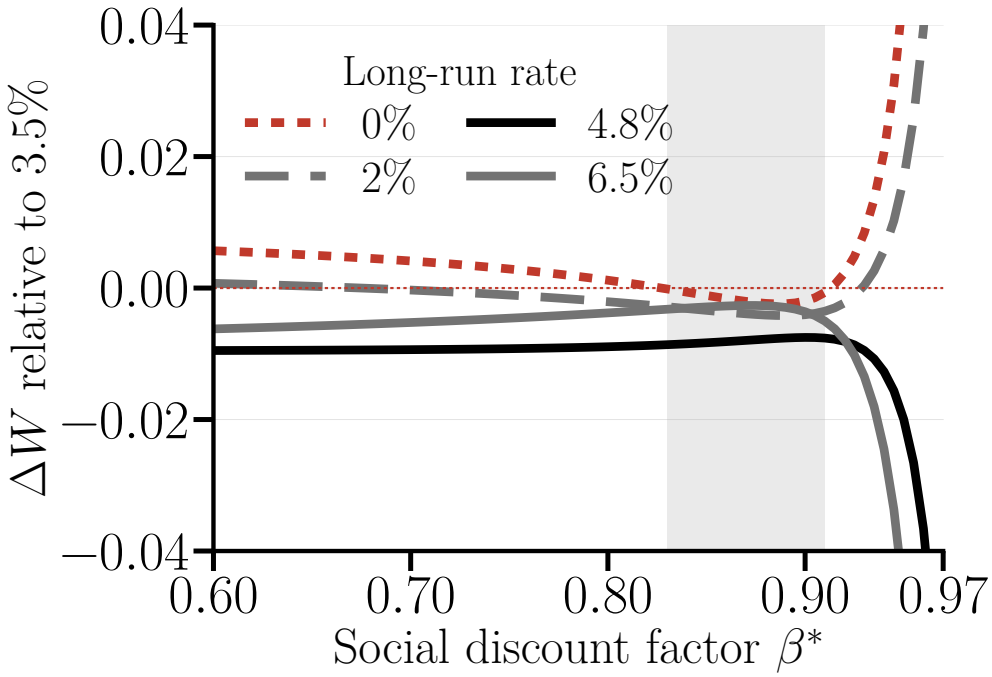


Figure C.6: Social welfare across long-run payroll rates

Notes. Each endpoint uses the main reform's dates and financing, with compensation re-solved. Each line is ΔW relative to the 3.5% rate, which is the dotted line at zero. Shading marks $\beta^* = 0.83$–0.91, where 3.5% leads among solved rates by a few thousandths of a utility unit; zero leads elsewhere.

D Robustness details

This appendix collects the tables and figures behind Section 6, in the same order. Each economy is scored against its own baseline, and in every variant the relief and the supplement are re-solved cohort by cohort until no cohort loses.

D.1 Learning by doing

Table D.1 reports recalibration, uncompensated outcomes, and the failed compensation attempts with fixed skills.

Table D.1: The same experiments with and without learning by doing

	Learning by doing	Fixed skills
Recalibrated household parameters		
Discount factor β, before growth adjustment	1.0099	1.0077
Leisure weight v	3.820	2.405
Learning efficiency η / depreciation δ_h	0.451 / 0.0471	0 / 0 (2010 profile fixed)
Uncompensated 3.5% schedule		
Losing cohorts (of 804)	35	100
Smallest cohort CEV (%)	-0.14	-0.74
Cohort born 2100: CEV (%)	$+3.47$	$+0.40$
Long-run cohort CEV (%)	$+7.73$	$+2.77$
ΔW, $\beta^* = 0.97$	$+0.62$	-0.85
Compensated transition, main instruments re-graded		
Losing cohorts after re-grading (main bands)	0	107
Losing cohorts after re-grading (bands to births 2090)	–	149
Smallest cohort CEV (%)	$+0.00008$	-3.06 (born 2046)
Uncompensated phase-out		
Losing cohorts (of 804)	51	94
Smallest cohort CEV (%)	-5.52	-4.41
Long-run cohort CEV (%)	$+10.52$	$+3.27$
Uncompensated 3.5% schedule in 2200, % relative to the baseline		
Output per adult	$+16.5$	$+10.8$
Hours per adult	$+8.8$	$+8.2$
Effective labor	$+13.4$	$+6.5$
Capital	$+22.9$	$+20.1$
Consumption per adult	$+15.7$	$+8.3$

Notes. The fixed-skill economy sets learning efficiency and skill depreciation to zero, and exogenous age multipliers preserve the main economy's 2010 mean skill profile. The discount factor and leisure weight are recalibrated to capital–output and hours targets, with bequest strength fixed and labor units matching the 2010 moment. Each economy uses its own baseline at the 150% ceiling. Compensation is re-solved for eight rounds with the main bands (supplement births 1977–2043; relief 2000–2045) and ten rounds with both extended to births 2090; fixed-skill rows with remaining losses are failed attempts. The uncompensated schedule keeps the main dates without either instrument, and the long-run block reports its outcomes in 2200. Fixed-skill transitions use a market tolerance of 10^{-4}.

D.2 Bequests and retirement resources

Table D.2 compares the main economy with the two economies without a bequest motive. Table D.3 reports the recalibrated parameters and the contribution of each cohort group to ΔW, and Figure D.1 the cohort profiles in the annuitized economy.

Table D.2: Results with and without a bequest motive

	Bequest motive		No bequest motive			
			Annuities		Inheritances	
	Main	Phase-out	Main	Phase-out	Main	Phase-out
Losing cohorts (of 804)	0	51	0	50	0	51
Smallest cohort CEV (%)	+0.00008	−5.52	+0.00190	−4.28	+0.00208	−6.45
ΔW, $\beta^* = 0$	+0.04	−0.54	+0.03	−0.36	+0.05	−0.54
ΔW, $\beta^* = 0.97$	+0.59	+3.09	+0.62	+3.85	+0.61	+3.93
Retention preferred for β^* up to	0.87		0.80		0.84	

Notes. Each economy is compared with its own debt-financed baseline under the 3.5% design, with compensation re-solved by cohort. Without a bequest motive $\vartheta = 0$, and the discount factor, leisure weight, learning efficiency, and skill depreciation are recalibrated to the same 2010 targets. Annuities: the assets of the deceased go to their cohort's survivors, and the retirement consumption floor stays. Inheritances: the assets of the deceased go to the living by recipient age, as in the main economy; compensation bands extend to births 2060 and the chain is verified at a market tolerance of 10^{-4}. The last row is the largest β^* at which the retained program beats the phase-out. ΔW is in utility units per 2026 adult.

Table D.3: Recalibrated parameters and welfare by cohort group

	Bequest motive		No bequest motive			
			Annuities		Inheritances	
	Main	Phase-out	Main	Phase-out	Main	Phase-out
Recalibrated household parameters						
Bequest strength ϑ	150.3		0.0		0.0	
Discount factor β, before growth adjustment	1.0099		1.0177		1.0200	
Leisure weight v	3.820		4.054		3.885	
Learning efficiency η	0.451		0.409		0.402	
Human-capital depreciation δ_h	0.0471		0.0416		0.0407	
Contribution to ΔW by cohort group, $\beta^* = 0.97$						
Alive in 2026, age 62 and over	+0.00	−0.59	+0.00	−0.47	+0.00	−0.67
Alive in 2026, under 62	+0.04	+0.04	+0.02	+0.11	+0.05	+0.13
Born 2007–2035	+0.04	+1.85	+0.02	+2.14	+0.03	+2.34
Born 2036–2100	+0.36	+1.54	+0.39	+1.79	+0.34	+1.84
Born after 2100	+0.16	+0.23	+0.18	+0.28	+0.19	+0.28
All cohorts	+0.59	+3.09	+0.62	+3.85	+0.61	+3.93

Notes. Upper block: recalibrated household parameters; other parameters and 2010 targets follow Section 3. Lower block: group contributions to ΔW relative to each economy's baseline, in utility units per 2026 adult at $\beta^* = 0.97$, summing to the total.

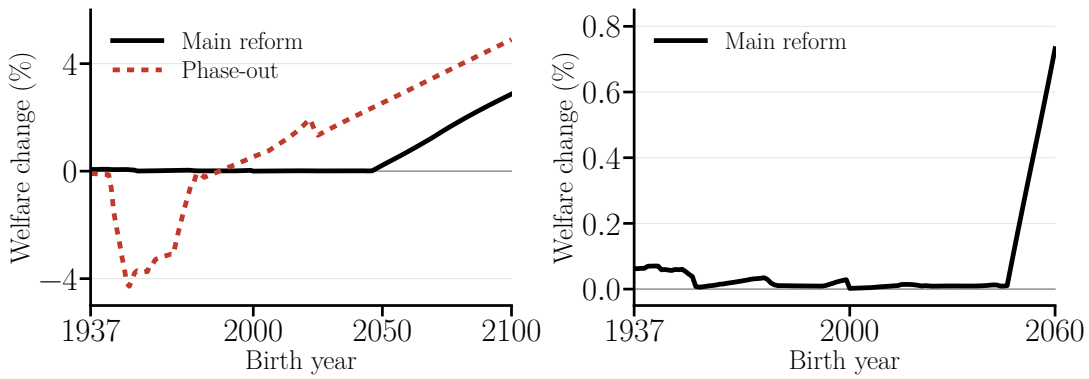


Figure D.1: Cohort welfare in the annuitized economy without a bequest motive

Notes. No-bequest economy with within-cohort asset pooling and the retirement consumption floor. CEVs are relative to that economy's baseline. Left: main reform and uncompensated phase-out, births 1937–2100. Right: main reform, births 1937–2060.

D.3 Compensation financing and adjustment speed

Table D.4 varies compensation financing and adjustment speed. Compensation is re-solved for the main-reform variants. The phase-out variants remain uncompensated.

Table D.4: Alternative compensation financing and adjustment speeds

Policy variant	Losing cohorts	ΔW, $\beta^* = 0$	ΔW, $\beta^* = 0.97$
Main reform	0	+0.04	+0.59
Relief financed by a surcharge on cohorts born 2044 onward	0	+0.04	+0.55
Slower main-reform adjustment, completed in 2300	0	+0.03	+0.34
Phase-out completed in 2200	51	−0.54	+3.09
Phase-out completed in 2300	42	−0.33	+2.19
Phase-out in 2200 with the baseline tax rule	51	−0.55	+3.04

Notes. All variants use the bequest economy and its debt-financed baseline. Counts cover 804 cohorts; ΔW is in utility units per 2026 adult. The market tolerance is 10^{-4} for the slower-adjustment case and 5×10^{-5} for the other rows.

D.4 Borrowing allowances

Table D.5 allows temporary borrowing above the main ceiling during compensation. Each path returns to a debt ratio of 150%.

Table D.5: Compensation financed under alternative debt ceilings

Borrowing allowance	Losing cohorts	$\Delta W, \beta^* = 0$	$\Delta W, \beta^* = 0.97$
150% of GDP (main reform)	0	$+0.04$	$+0.59$
152% of GDP while compensation is paid	0	$+0.07$	$+0.61$
160% of GDP while compensation is paid	0	$+0.12$	$+0.66$

Notes. Debt may exceed 150% of GDP while compensation is paid and returns to 150% afterward; instruments otherwise follow the main reform. All variants use the same debt-financed baseline. Counts cover 804 cohorts; ΔW is in utility units per 2026 adult.

D.5 Protecting every household

Table D.6 compares the household-level Pareto path of Section 6.5 with the main reform.

The household-level path is valued node by node with the same tool as the main reform. The last two columns of Table B.2 report its mean and worst CEV for each group; no household in any group loses. The search that produced it keeps every numerical limit of the main reform (market residual 5e-5, budget and payroll identities at 10^{-8}) and records an independent replay of the welfare, market, budget, and payroll checks; its cohort tables and exported paths are the inputs to Table D.6.

Table D.6: The main reform and the household-level Pareto path

	Main reform	Household-level path
Households		
Adults alive in 2026 who lose (%)	4.3	0.0
Worst household CEV, adults alive in 2026 (%)	-0.170	$+0.001$
Worst household CEV, entrants 2030–2200 (%)	-0.029	$+0.001$
Cohort averages		
Losing cohorts (of 804)	0	0
Cohort born 2100: CEV (%)	$+3.42$	$+3.36$
Long-run cohort CEV (%)	$+7.73$	$+7.73$
ΔW, $\beta^* = 0$	$+0.04$	$+0.05$
ΔW, $\beta^* = 0.97$	$+0.59$	$+0.53$
Instruments		
Payroll relief: birth cohorts	2001–2042	2000–2050
Payroll relief: largest (percentage points)	2.29	2.60
Benefit supplement: birth cohorts	1981–2040	1937–2050
Benefit supplement: largest (% of paid benefit)	8.6	22.0
Benefit supplement: first payment	2090 or age 75 (from retirement if born before 2000)	2026 for cohorts already retired; otherwise as in the main reform
Financing		
Peak Treasury transfer (% of GDP)	0.46 (2065)	0.65 (2090)
Present value of transfers (% of 2026 GDP)	2.15	3.30
Consumption tax above baseline: at most (points)	0.45	0.75
Consumption tax below baseline from	2072	2091

Notes. Both designs use the 3.5% schedule, the 150% ceiling, and the debt-financed baseline. Household rows come from Table B.2: a household loses if its CEV is below -0.01%, and entrants are the cohorts born 2030–2200. The household-level path re-solves relief and supplement on each cohort's worst occupied node until no node in the 804 cohorts is below zero (worst $+0.0005\%$, born 2049), verified by an independent replay at a market tolerance of 5×10^{-5}. Instrument sizes are table values by birth year; relief cohorts have at least 0.05 points. Present values use each path's own interest rates.